The Frinkahedron

Geometry, Algebra, and Topology

Bill Cochran

A monograph on enclosure, symmetry,
and recoverable observation

October 2026

"What if the universe only *looks like* it is expanding?"

Preface

A description of a whole is useful only when its relation to the observations of that whole can be stated. An enclosing surface, a sequence of positions on a clock, and a record of completed loops may describe the same admitted configuration, but each presents a different recovery problem. The purpose of this monograph is to construct one setting in which those problems can be posed and solved together.

The object called the frinkahedron is realized here as the boundary of a product of two regular hexagons. This boundary is a three-sphere. A specified homeomorphism carries its two solid-torus pieces to the standard decomposition of the round sphere, where two families of great circles admit both finite permutation descriptions and continuous quaternion descriptions. Projection then becomes a mathematical operation with a calculable loss of information. The same approach extends to period records on an explicit cell complex and to bounds for their reconstruction.

The exposition is intended for readers with some linear algebra, elementary analysis, and familiarity with groups and continuous maps. The required algebraic topology is introduced through the particular spaces and matrices used in the construction. Proofs of the model-specific assertions are included; references locate the standard theory surrounding them. In particular, the finite example is developed far enough to recover every one of its thirty-six states and to compute its error margins.

The opening question concerns appearance and inference. The mathematical response developed here is to identify an observation map, determine its fibers, and retain the additional data required to distinguish their members. The resulting sphere is a configuration space with a positive definite metric. A physical interpretation would require a calibrated relation between these coordinates and physical observables. The present work establishes the geometric construction and its observation theory on their stated domains.

This monograph grew from the geometric specification of the accompanying *Measurement* project. The episode manuscripts and the earlier `instrument/` study supply its questions about enclosure, counting, closure, and retained readings. Their relation to the construction is discussed in Appendix A. The companion tolerances manuscript develops the variational and numerical models. Here the organizing question is how a whole can admit several descriptions while the information needed for recovery remains explicit.

Notation and conventions

All spaces, maps, and norms are specified when introduced. The following notation collects the principal objects used throughout the monograph.

Symbol	Meaning
P_6, E_j	Regular unit-circumradius hexagon and its j-th edge.
K, F	Product $P_6 \times P_6 \subset \mathbb{R}^4$ and its boundary $F = \partial K$.
$S^3, \overline{B}^3$	Unit sphere in $\mathbb{R}^4$ and closed unit ball in $\mathbb{R}^3$.
Ψ	The specified homeomorphism $F \to S^3$.
$z = (z_1, z_2)$	Complex coordinates on $\mathbb{R}^4 \cong \mathbb{C}^2$.
η, α, β	Latitude and phase coordinates: $z = (\cos\eta\, e^{i\alpha}, \sin\eta\, e^{i\beta})$.
P, L_σ	Orthogonal observation and its sheet-labelled inverse.
C_+, C_-	Simultaneous-phase and opposite-phase circle actions.
H_+, H_-	The corresponding Hopf observations into S^2.
$T_{\pi/4}$	Clifford torus, where $\lvert z_1\rvert = \lvert z_2\rvert = 1/\sqrt{2}$.
$\mathcal{D}, D, E$	General admitted state set, finite clock state set, and state encoding.
O, R	Observation and retained lifting data, also called a receipt.
d_0, d_1	Coboundary maps of the measured cell complex.
$\Delta_1, \mathcal{H}^1$	Hodge Laplacian on one-cochains and its kernel.
λ, γ	Positive spectral gap and smallest observable harmonic response.
M	Measurement map used to recover the harmonic component.
θ	Six-position clock step $\pi/3$.
S_+, S_-	Finite steps $(j,k) \mapsto (j+1, k \pm 1)$ modulo six.

Euclidean vector norms and their induced operator norms are used unless another inner product is declared. The round geodesic metric and Euclidean chord distance on S^3 are distinguished. Angles are in radians; phase coordinates are understood modulo 2π. A bar denotes complex conjugation, and a star denotes the adjoint for the stated inner products. Integer subscripts on hexagon vertices and edges are understood modulo six.

A *calibrated* description specifies the coordinates, inner products, scales, and observation maps needed for its comparisons. Deterministic error bounds are used throughout the worked realization. A probability distribution is introduced only for the separate information counts in Chapter 4.

Contents

Preface **i**

Notation and conventions **ii**

1 The problem of recoverable observation **1**
1.1 A whole and its observable descriptions . . . 1
1.2 The basic diagram . . . 1
1.3 The construction and its principal results . . . 2
1.4 Geometric, algebraic, and topological descriptions . . . 3
1.5 Relation to the literature and organization . . . 3

2 Enclosure and the geometry of projection **4**
2.1 The product enclosure and its cells . . . 4
2.2 From the polygonal boundary to the three-sphere . . . 6
2.3 Intrinsic distance and ambient interior . . . 7
2.4 Orthogonal observation and its inverse . . . 8
2.5 Stereographic coordinates and apparent scale . . . 9

3 Circle actions, winding, and algebraic representation **11**
3.1 The torus coordinates and two great-circle families . . . 11
3.2 Hopf observations and their fibers . . . 12
3.3 The paired reading and its antipodal ambiguity . . . 12
3.4 Quaternion multiplication and the rotation group . . . 14
3.5 Finite clocks and exact intertwining . . . 15
3.6 The sixfold symmetry and its two sectors . . . 15

4 Observation, cohomology, and retained information **17**
4.1 Enclosure and accessible domains . . . 17
4.2 Observation fibers and the recovery problem . . . 18
4.3 Finite ambiguity and approximate readings . . . 19
4.4 A cellular model of the accessible torus . . . 20
4.5 Harmonic representatives and measured periods . . . 21
4.6 Transport of representation and metric . . . 22

5 Stability of observation and reconstruction **23**
5.1 Residual energy and the harmonic component . . . 23
5.2 Period measurements and uniqueness . . . 24
5.3 The torus calculation in full . . . 25
5.4 Perturbations and calibration margins . . . 26
5.5 Composition of operations and the role of order . . . 26
5.6 Comparison through a common observable . . . 27
5.7 Trace errors as inputs to reconstruction . . . 28

6 A finite realization and its reconstruction **29**
6.1 States, encodings, and the action of the clocks . . . 29
6.2 Orthogonal observation and the retained sheet . . . 30
6.3 Paired circle observations and their antipodal fiber . . . 31
6.4 Separation and finite decoding . . . 31
6.5 A worked state and a deterministic error bound . . . 32
6.6 A calibrated period record . . . 33
6.7 The combined reconstruction . . . 34

7 Enclosure and recoverability **35**
7.1 The completed correspondence . . . 35
7.2 Exact structure and quantitative response . . . 36
7.3 Extensions of the construction . . . 36

A Sources and the episode correspondence **38**
A.1 Origin of the construction . . . 38
A.2 Episode concordance . . . 38
A.3 The earlier observation framework . . . 39
A.4 Mathematical construction and source evidence . . . 39

Bibliography **40**

Chapter 1

The problem of recoverable observation

1.1 A whole and its observable descriptions

An object may be represented by a surface, by a list of coordinates, or by a rule for moving between configurations. These descriptions support different operations. A surface can enclose; a list can be permuted; a motion can be composed with its inverse. To compare them mathematically, one must give the maps connecting their admitted states and explain which observations survive the passage from one description to another.

The question becomes especially clear for a three-sphere. Its orthogonal image in three-dimensional space is a filled ball. An interior point of that ball has two preimages on the sphere, while a point of its boundary has one. The image encloses a region, yet its coordinates alone do not recover the original point. This elementary example already separates three matters: the topology of the whole, the shape of its observation, and the information needed to lift that observation back to a state.

This monograph develops the separation through one explicit construction. The whole is a polyhedral boundary $F \subset \mathbb{R}^4$, subsequently identified with the round sphere. Its motions include two circle actions. Its finite realization consists of two six-position clocks. A second record, carried by a measured torus complex, makes local closure and global period recovery available for direct calculation. Each description has its own observation map, and the comparison between them retains the data on which recovery depends.

The expression *the whole* will therefore have a precise domain. For the continuous geometry it means all of F, or equivalently all of S^3. For the finite realization it means all thirty-six states in the admitted clock sector. A statement quantified over that sector is complete on its specified domain. Passing to a larger state set requires a new encoding and a corresponding recovery argument.

1.2 The basic diagram

Let $\mathcal{D}$ be a set of admitted states, let X be a representation space, and let $E : \mathcal{D} \to X$ encode those states. An observation is a map $O : X \to Y$. Its value $O(E(s))$ is the reported datum. Additional retained information is represented by a map $R : \mathcal{D} \to \mathcal{A}$. The problem of exact recovery is to construct a reader $\mathcal{L}$ such that

$$\mathcal{L}\big(O(E(s)), R(s)\big) = s, \qquad s \in \mathcal{D}. \tag{1.1}$$

Only pairs in the image of $(O \circ E, R)$ need have an assigned recovered state. A practical extension to nearby data is a further problem of stability, which requires metrics and a specified reconstruction domain.

A necessary and sufficient set-theoretic condition for (1.1) is that $(O \circ E, R)$ be injective. Necessity follows by applying the reader to two equal data pairs. Conversely, an injective map has a unique inverse on its image. This elementary fact sets the agenda for the whole construction: determine the fibers of the observation, then supply lifting data that separate their members.

The word *receipt* will occasionally denote these retained lifting data. It may be a sheet sign, a branch label, a permutation, or an integer retained beside a quotient class. Its mathematical role is always the same: it is an explicit argument of the inverse map. The size of a receipt depends on the observation. A single Hopf observation forgets an entire circle, whereas the pair of Hopf observations considered below forgets only an antipodal choice away from two exceptional core circles.

For a desired observable $q : \mathcal{D} \to V$, full state recovery can be replaced by recovery of $q(s)$. In that case the relevant condition is that q be constant on every fiber of the retained data map. This permits an observation to be sufficient for one question while leaving other state coordinates undetermined. It also explains why agreement of two computed readings is a statement about their common observable and encoding assumptions. The geometry and the measurement model determine the meaning of that agreement [9].

1.3 The construction and its principal results

The selected polytope is

$$K = P_6 \times P_6 \subset \mathbb{R}^2 \times \mathbb{R}^2, \qquad F = \partial K,$$

where P_6 is the regular hexagon of circumradius one. Its boundary has thirty-six vertices, seventy-two edges, forty-eight two-dimensional faces, and twelve three-dimensional facets. Each facet is a hexagonal prism. These cells give a concrete meaning to the enclosure before any projection is drawn.

The first principal result is an explicit homeomorphism $\Psi : F \to S^3$. The map is chosen with more structure than a bare radial identification: it sends the common boundary of the two product solid tori to the round Clifford torus. This makes the polyhedral construction compatible with the circle coordinates used in the algebraic chapters. The round metric can then be transported to F, so both the topology and the metric used for the actions are declared.

The second result concerns observation. Orthogonal projection has a two-sheet inverse in the interior of its ball image. A retained sheet label gives a Lipschitz inverse on every domain with a positive distance from the silhouette. The paired Hopf observation has a different twofold ambiguity: it identifies z with $-z$ away from the core circles. The two circle directions have winding vectors $(1, 1)$ and $(1, -1)$, whose determinant is -2. The same index-two phenomenon appears in the finite six-position clocks, where the two diagonal steps generate two sectors of eighteen states each.

The third result makes closure quantitative. On the explicit torus cell complex, the one-cochain Hodge Laplacian has a two-dimensional kernel and positive spectral gap six. A two-coordinate measurement has smallest harmonic response $1/\sqrt{3}$. These constants turn a local residual and a period perturbation into separate reconstruction bounds. The framework also gives operator perturbation estimates and an error bound for a retained word of operations.

The final result combines these constructions in a complete finite example. The thirty-six clock states have twenty-four distinct orthogonal observations and eighteen distinct paired Hopf observations. Every fiber is enumerated and inverted using its specified receipt. Separation distances give exact thresholds for discrete decoding. A worked observation with declared bounded errors

then illustrates how geometric and cochain estimates enter one common comparison. Its numbers are derived from the selected model; the example requires no unstated measurement record.

The contribution is the explicit assembly and comparison of these structures. The component theories—spheres, Hopf fibrations, Clifford actions, finite cochain complexes, and singular-value estimates—are standard. Their role here is to make the correspondence between enclosure, motion, counting, and recovery calculable on the same admitted states.

1.4 Geometric, algebraic, and topological descriptions

Geometry supplies the state space, its metric, and its observation maps. The same topological sphere supports many metrics, and the same sphere admits many maps into three-dimensional space. For this reason an orthogonal projection and a stereographic chart are treated separately. The former has a compact ball image and loses a sheet; the latter gives coordinates on the sphere with one point omitted. Neither drawing can replace the definition of the map that produced it.

Algebra supplies the operations. The cyclic permutations of a sampled orbit and the quaternion motion through its samples satisfy an exact intertwining identity. This identity states how an operation in one description is carried to the other. An arbitrary permutation need not arise from the chosen motion, and a continuous motion need not preserve a polyhedral cell decomposition. The relevant subgroup and its action must therefore be identified before an operation is transferred.

Topology supplies the relations that persist under continuous changes of coordinates. A loop on a torus may represent a nonzero homology class while being contractible in the containing three-sphere. Similarly, deleting a core circle changes the accessible region and the allowed contractions. The space on which a period is defined forms part of the claim about that period. Cellular topology makes these distinctions visible in finite incidence maps; positive inner products then give those maps a quantitative Hodge theory [6, 7, 1].

The three descriptions meet in reconstruction. A topological calculation identifies the undetermined harmonic directions. A calibrated algebraic map measures those directions. Geometry and linear analysis supply the constants that control how errors propagate. Exact identities specify the model; positive lower bounds make its inverse stable. These are successive parts of one argument.

1.5 Relation to the literature and organization

The geometric discussion uses the sphere, its product decomposition, and the Hopf construction as presented in the standard topology literature and in explicit geometric accounts [6, 10, 11]. The quaternion and Clifford descriptions follow declared sign conventions consistent with the treatments of Woit and Gallier [12, 4]. The cochain discussion uses finite Hodge theory and compatible transport of complexes; the surrounding numerical framework is developed in combinatorial Hodge theory and finite element exterior calculus [7, 1]. Singular values provide the finite-dimensional inverse estimates [5].

Chapter 2 constructs the polyhedral whole and studies its projections. Chapter 3 develops its two selected circle actions, their Hopf observations, and their finite and Clifford descriptions. Chapter 4 relates observation fibers to lifting data and locates the periods in an explicit complex. Chapter 5 proves reconstruction and stability estimates. Chapter 6 works through the finite realization. The concluding chapter draws together the exact comparison and its remaining extensions. The appendix records source provenance separately from the mathematical development.

Chapter 2

Enclosure and the geometry of projection

The geometric object studied in this monograph is a boundary together with its possible observations. This choice makes the relation between a whole and a partial reading explicit. The whole has an intrinsic dimension, an embedding in a larger space, and a collection of maps by which it can be represented. Those three structures determine different questions. An embedding determines where the object lies; an observation determines which distinctions survive; a metric determines how a perturbation is measured.

We begin with a concrete polytope rather than an unspecified picture of a higher-dimensional enclosure. The construction supplies twelve facets, their attaching data, and a preferred torus separating the boundary into two regions. An explicit homeomorphism then carries this boundary onto the round three-sphere. The homeomorphism is chosen to preserve the separating torus, so that the cellular construction and the circle geometry of the next chapter can be compared by formulas.

The four ambient coordinates used here are configuration coordinates. Their mathematical role is to support a Euclidean realization of the boundary and its symmetries. Any subsequent physical interpretation must identify the observable to which those coordinates are calibrated. The geometric results themselves concern the declared configuration space.

2.1 The product enclosure and its cells

Put $\theta = \pi/3$, and let

$$u_j = (\cos j\theta, \sin j\theta), \qquad P_6 = \operatorname{conv}\{u_0, \dots, u_5\} \subset \mathbb{R}^2, \qquad K = P_6 \times P_6 \subset \mathbb{R}^4.$$

All vertex and edge subscripts are taken modulo six. The regular hexagon contains the origin in its interior. With

$$c = \cos(\pi/6) = \sqrt{3}/2, \qquad n_j = (\cos((j+\tfrac{1}{2})\theta), \sin((j+\tfrac{1}{2})\theta)),$$

its supporting-half-plane description is

$$P_6 = \{u \in \mathbb{R}^2 : n_j \cdot u \le c \text{ for all } j\}.$$

The edge E_j joining u_j to u_{j+1} lies in the supporting line $n_j \cdot u = c$. Consequently the twelve inequalities for K separate into six in each coordinate plane. This separation will account both for the cell census and for the two solid tori in the boundary.

Definition 2.1. The *frinkahedron* of the present construction is the polyhedral boundary

$$F = \partial(P_6 \times P_6).$$

Its facets are its three-dimensional cells. A point of F is a complete boundary configuration, including both coordinate-plane components.

The vertices of K are the 36 ordered pairs (u_j, u_k). Its facets are

$$A_j = E_j \times P_6, \qquad B_k = P_6 \times E_k.$$

Each facet is a hexagonal prism. Within one family, adjacent edges of the underlying hexagon give adjacent facets: $A_j \cap A_{j+1} = \{u_{j+1}\} \times P_6$, a hexagonal two-cell. Nonadjacent edges give disjoint facets. The same rule holds for the B family. Across the families, every pair meets, and

$$A_j \cap B_k = E_j \times E_k$$

is a square. Intersections of lower dimension are obtained by intersecting the two factors separately. Thus the attaching information is determined by the product, without an appeal to hidden edges in a perspective drawing.

Proposition 2.2. *The boundary F has f-vector $(36, 72, 48, 12)$. Every two-dimensional face belongs to exactly two facets, and the oriented sum of the facets is a three-cycle.*

Proof. A face of a product of polytopes is the product of a face in each factor. The proper faces here have dimensions adding to at most three. There are $6 \cdot 6 = 36$ vertices. An edge is an edge in one factor times a vertex in the other, giving $2 \cdot 6 \cdot 6 = 72$. Two-faces are the 36 edge products and the twelve vertex-times-hexagon products. The twelve facets have already been enumerated.

A square $E_j \times E_k$ belongs to A_j and B_k. A hexagon $\{u_j\} \times P_6$ belongs to the two A facets corresponding to edges incident on u_j; the other kind of hexagon has the analogous two B facets. Orient $\mathbb{R}^4$ by its ordered coordinate planes, and give each facet its boundary orientation. The outward normals are $(n_j, 0)$ and $(0, n_k)$. At a shared two-face the two induced boundary orientations are opposite. Each term therefore cancels in the boundary of the sum. □

Dimension	Cell types	Count	Signed count
0	vertex × vertex	36	+36
1	edge × vertex, and reverse	72	−72
2	squares; hexagons in either factor	36 + 12	+48
3	edge × hexagon, and reverse	12	−12

Table 2.1: The cells of F. The Euler characteristic is $36 - 72 + 48 - 12 = 0$. Two-dimensional faces and three-dimensional facets enter different parts of the census.

The cycle in the proposition is the fundamental boundary cycle. Its cancellation expresses enclosure at the level of incidence: a two-face is shared with opposite orientations, so the union has no exposed two-dimensional boundary. The alternating count is a useful consistency relation, but it does not by itself identify a space. Many spaces have Euler characteristic zero. The sphere topology will follow from the homeomorphism constructed below.

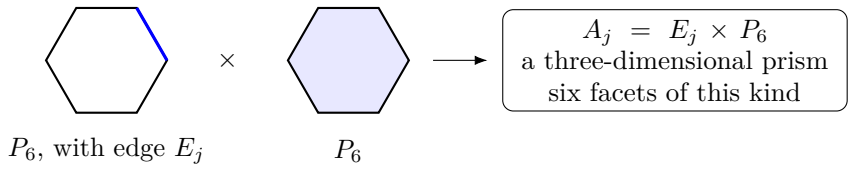


Figure 2.1: A product incidence diagram. Interchanging the factors gives the other six facets. The diagram records a construction of cells rather than a perspective image of the four-polytope.

Example 2.3 (Dimension and incidence)**.** A simple parity test illustrates why a cell count includes its dimensions and attaching rules. In a closed polygonal two-manifold, every edge occurs in exactly two polygon boundaries, so the total number of panel sides must be even. Three quadrilaterals and nine triangles contribute $3 \cdot 4 + 9 \cdot 3 = 39$ sides and cannot form such a manifold with these edge incidences. The twelve facets of F are three-dimensional prisms; their two-dimensional faces are the 48 entries in Table 2.1. The count twelve acquires a geometric meaning through this specified dimension and incidence structure.

2.2 From the polygonal boundary to the three-sphere

A convex body containing the origin can be described by its distance to the boundary in each radial direction. For a unit vector $v \in S^1$, that distance for P_6 is

$$r(v) = \frac{c}{\max_j n_j \cdot v}.$$

The six normals are equally spaced. One lies within angle $\pi/6$ of any given direction, and therefore

$$c \le \max_j n_j \cdot v \le 1, \qquad c \le r(v) \le 1.$$

These bounds make the radial change of coordinates well behaved even at the origin.

Lemma 2.4. *The map $g : P_6 \to D^2$ defined by $g(0) = 0$ and*

$$g(u) = \frac{u}{r(u/\|u\|)} \quad (u \neq 0)$$

is a homeomorphism carrying ∂P_6 onto S^1. It commutes with rotation through θ.

Proof. For a fixed unit direction v, every point of the polygon on that ray has the unique form tv, $0 \le t \le r(v)$. The formula sends it to $(t/r(v))v$. Its inverse on the disk is consequently

$$g^{-1}(sv) = sr(v)v, \qquad 0 \le s \le 1.$$

The radius function is continuous and positive. The displayed uniform bounds prove continuity of both maps at zero; away from zero continuity follows from the formulas. Rotation through θ permutes the normals, leaving their maximum unchanged. Hence r has that rotational symmetry, and so does g. □

Let $G = g \times g : K \to D^2 \times D^2$, identifying each disk with $\mathbb{C}$. It carries the product boundary onto

$$\partial(D^2 \times D^2) = \{(v_1, v_2) : \max(|v_1|, |v_2|) = 1\}.$$

There are two successive normalizations: each polygon is first carried to a disk, and then the product boundary is carried to a Euclidean sphere. Define

$$\Psi : F \longrightarrow S^3, \qquad \Psi(p) = \frac{G(p)}{\|G(p)\|}. \tag{2.1}$$

Proposition 2.5. *The map Ψ is a homeomorphism. It carries the product seam $\partial P_6 \times \partial P_6$ onto the Clifford torus*

$$T = \{(z_1, z_2) \in S^3 : |z_1| = |z_2| = 1/\sqrt{2}\}.$$

Proof. For $z \in S^3$, let $m(z) = \max(|z_1|, |z_2|)$. This number is positive, and $z/m(z)$ belongs to the product boundary. Direct substitution gives the inverse

$$\Psi^{-1}(z) = G^{-1}\left(\frac{z}{m(z)}\right).$$

Both maps are continuous. On the product seam, the two components of $G(p)$ both have norm one, so normalization gives two component norms $1/\sqrt{2}$. Conversely, equality of the component norms of z forces both components of $z/m(z)$ onto the unit circles. □

The choice of Ψ matters to the seam statement. Raw radial normalization $p \mapsto p/\|p\|$ is also a homeomorphism from F onto S^3, because every ray meets a convex boundary exactly once. Along the polygon seam, however, the component radii generally differ. Raw normalization therefore need not carry that seam onto the round Clifford torus. The factorwise map G supplies the missing equality before the final normalization.

The two halves of the boundary are

$$A = \partial P_6 \times P_6, \qquad B = P_6 \times \partial P_6, \qquad F = A \cup B.$$

Each is a solid torus, since $\partial P_6 \cong S^1$ and $P_6 \cong D^2$. Their common boundary is the seam. Under Ψ, the two halves become

$$\Psi(A) = \{|z_1| \geq |z_2|\}, \qquad \Psi(B) = \{|z_2| \geq |z_1|\}.$$

The first retracts to the core $(e^{i\alpha}, 0)$ and the second to $(0, e^{i\beta})$. A contraction of the second component in the first region, followed by normalization, gives the first retraction explicitly; the coordinate interchange gives the other. This realizes the familiar decomposition of S^3 into two solid tori [11, §3.2] with the polygonal attaching data retained.

2.3 Intrinsic distance and ambient interior

Topology identifies which configurations are adjacent in a continuous description. Quantitative comparisons also require a metric. Throughout the monograph the round metric is transported through Ψ:

$$d_F(p, q) = \arccos\big(\Psi(p) \cdot \Psi(q)\big).$$

For state errors we use the Euclidean chord norm in the normalized embedding. If $z = \Psi(p)$ and $z' = \Psi(q)$, then

$$\|z - z'\| = 2\sin\left(\frac{d_F(p, q)}{2}\right).$$

Thus infinitesimal chord and round distances agree, while a global conversion is explicitly available. These distances differ from the piecewise flat length metric obtained by treating each prism facet

as Euclidean. A great circle transported back to F can pass through facet interiors and need not follow a polyhedral edge.

The choice separates three uses of dimension. The configuration F has intrinsic dimension three: it is homeomorphic to S^3. Its ambient realization uses four coordinates. The body K that it bounds has four-dimensional interior. The boundary itself has empty interior as a subset of $\mathbb{R}^4$; every neighborhood of a boundary point also contains points outside the boundary. Equivalently for the normalized model, the equation $\|z\| = 1$ defines a three-dimensional boundary, while $\|z\| < 1$ defines the bounded open four-ball.

Accordingly, saying that the frinkahedron has no four-dimensional interior refers to the boundary as a set. It remains the boundary of a four-dimensional filling. The filling is useful for describing enclosure, but its points are not additional boundary configurations. Likewise, overlap in a projected image identifies observation fibers; it does not attach distinct facets to one another. This distinction is essential when one asks how a three-dimensional image can appear to enclose a volume.

2.4 Orthogonal observation and its inverse

Write a normalized point as $z = (x, w) \in \mathbb{R}^3 \times \mathbb{R}$, and let $P(z) = x$. Solving the sphere equation gives

$$P(S^3) = \overline{B}^3, \qquad L_\sigma(x) = \left(x, \sigma\sqrt{1 - \|x\|^2}\right), \quad \sigma \in \{+1, -1\}. \tag{2.2}$$

Every interior point of the image has two representatives, and every point of its boundary sphere has one. The observed image is therefore a filled three-ball, although the observed object is a three-sphere.

One may reconstruct the topology by taking two copies of $\overline{B}^3$ and identifying their boundary spheres pointwise. The map induced by $(x, \sigma) \mapsto L_\sigma(x)$ is a continuous bijection onto S^3. The source quotient is compact and the sphere is Hausdorff, so it is a homeomorphism. The two signs distinguish the sheets over the open ball; on the silhouette they represent the same point. We call a retained sign a *sheet receipt*, meaning auxiliary information specifying a representative of an observation fiber.

Proposition 2.6. *Projection is nonexpansive in chord norm. On either fixed sheet, restricted to* $\|x\| \le \sqrt{1 - a^2}$ *with* $0 < a \le 1$*, its inverse lift is* a^{-1}*-Lipschitz.*

Proof. Deleting a coordinate cannot increase Euclidean distance. For $\|x\| < 1$, differentiation of the lift gives

$$DL_\sigma(x)^T DL_\sigma(x) = I + \frac{xx^T}{1 - \|x\|^2}.$$

The derivative preserves lengths in directions orthogonal to x. In the radial direction its squared stretching factor is $(1 - \|x\|^2)^{-1}$. Hence its operator norm is at most $1/a$ on the stated ball. This ball is convex, so integration along the line segment between two observations proves the finite-distance bound. □

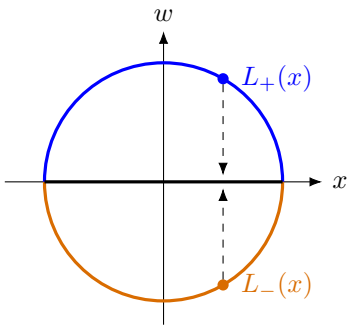
The margin a is a lower bound on the magnitude of the omitted coordinate. It quantifies the separation from the silhouette, where radial motion becomes difficult to reconstruct. The loss is intrinsic to this observation map. Taking $x_r = re_1$ and $x_1 = e_1$ on one sheet gives

$$\|x_1 - x_r\| = 1 - r, \qquad \|L_\sigma(x_1) - L_\sigma(x_r)\| = \sqrt{2(1 - r)}.$$

Their ratio diverges as $r \uparrow 1$, so no uniform Lipschitz inverse exists on the whole closed ball. Nevertheless the lift remains continuous. Indeed, with $d = \|x - y\| \le 2$, the inequality $\left|\sqrt{s} - \sqrt{t}\right|^2 \le |s - t|$ yields

$$\|L_\sigma(x) - L_\sigma(y)\|^2 \le d^2 + 2d \le 4d.$$

The inverse is globally Hölder continuous with exponent $1/2$, and the radial example shows why that exponent occurs at the silhouette.



A two-dimensional section. In the full map, the interval is a filled B^3 and the endpoints are a silhouette S^2.

Figure 2.2: Orthogonal projection forgets the sheet. The representatives coincide on the silhouette, where the inverse changes from Lipschitz behavior on interior subballs to square-root behavior.

The normalized observation is the composite $P \circ \Psi$ on F. Direct linear projection of the original polytope boundary is a different map. For any projection deleting one coordinate, $P(F) = P(K)$: a nonempty line section of the compact convex body has boundary endpoints, and those endpoints retain the same projected coordinates. Its image is the projection of the polytope, rather than the round ball obtained after normalization. Choosing an image therefore includes choosing whether the coordinate change precedes the projection.

2.5 Stereographic coordinates and apparent scale

There is a second useful three-dimensional representation. Stereographic projection from $N = (0, 0, 0, 1)$ is

$$\Sigma(x, w) = \frac{x}{1 - w}, \qquad \Sigma^{-1}(y) = \left(\frac{2y}{1 + \|y\|^2}, \frac{\|y\|^2 - 1}{1 + \|y\|^2}\right).$$

Substitution proves that these are inverse maps between $S^3 \setminus \{N\}$ and all of $\mathbb{R}^3$. The chart is particularly useful for displaying circles and their linking [11, §1]. It has no compact silhouette and forgets no sheet on its domain. Instead it omits one point of the sphere, whose approach corresponds to arbitrarily large chart coordinates.

The scale change can be calculated rather than inferred from a drawing. Differentiating Σ^{-1} and taking the Euclidean inner product in $\mathbb{R}^4$ gives the round line element

$$ds^2 = \frac{4\,\|\,\mathrm{d}y\|^2}{(1 + \|y\|^2)^2}.$$

The chart preserves angles, while a unit displacement in chart coordinates has round length $2/(1 + \|y\|^2)$. Conversely, its local magnification relative to round length is $(1 + \|y\|^2)/2 = 1/(1 - w)$. That factor becomes unbounded near the omitted pole.

An apparent enlargement in this representation can therefore arise from the varying scale of the chart. To treat such an enlargement as a change of physical distance requires an observable and a calibration connecting the chart to that observable. This is the mathematical content of asking what a representation makes visible: an image can faithfully display a point and its neighborhood while displaying their scale nonuniformly. The following chapter supplies motions on the whole sphere, so that both their geometry and their representation can be studied within the same construction.

2.5.1 The volume of a whole and the volume of its image

The line element also determines how volume is represented. In three chart dimensions its conformal factor gives

$$\mathrm{d}V_{S^3} = \frac{8}{(1+\|y\|^2)^3}\,\mathrm{d}y, \qquad \mathrm{Vol}(S^3) = 32\pi \int_0^\infty \frac{r^2}{(1+r^2)^3}\,\mathrm{d}r = 2\pi^2.$$

For the integral, substitute $r = \tan t$; the remaining integral of $\sin^2 t \cos^2 t$ from zero to $\pi/2$ equals $\pi/16$. The unbounded chart therefore represents a sphere of finite intrinsic volume.

Orthogonal observation yields a complementary calculation. The graph metric in Proposition 2.6 has determinant $(1-\|x\|^2)^{-1}$. Hence a single sheet has volume element

$$\mathrm{d}V_{\mathrm{sheet}} = \frac{\mathrm{d}x}{\sqrt{1-\|x\|^2}}, \qquad \mathrm{Vol}(\mathrm{sheet}) = 4\pi \int_0^1 \frac{r^2}{\sqrt{1-r^2}}\,\mathrm{d}r = \pi^2.$$

Here $r = \sin t$ reduces the integral to that of $\sin^2 t$. The two sheets recover total volume $2\pi^2$; their common silhouette has zero three-dimensional volume. The image ball itself has Euclidean volume $4\pi/3$. These quantities use different metrics and count different fibers.

More precisely, pushing the round volume through P gives density $2/\sqrt{1-\|x\|^2}$ on the open ball. The density is unbounded at the silhouette but integrable. It records both the stretching of a sheet inverse and the multiplicity of the observation. These calculations concern a fixed geometric object. They make explicit how an image can change apparent scale or volume with its chosen representation, while inference about a changing physical volume still requires a calibrated distance and observation model.

Chapter 3

Circle actions, winding, and algebraic representation

The sphere model provides a setting in which a motion can be described geometrically, algebraically, or by a finite clock. The point of comparison is the action on configurations. A permutation describes a sampled motion when an embedding intertwines the permutation with that motion; quaternion multiplication describes it when the multiplication yields the same coordinate map. The common action supplies the mathematical identity among the descriptions.

Two families of great circles will organize this comparison. On the separating Clifford torus they run along the two diagonal directions of an angular square. Their winding vectors span a real plane but an index-two integer lattice. The resulting distinction between real coordinates and integer windings reappears as a two-state ambiguity of a paired observation and as a parity restriction in the finite clock. We develop those three appearances from the same phase formulas.

3.1 The torus coordinates and two great-circle families

Use complex coordinates on the normalized sphere:

$$(z_1, z_2) = (\cos\eta\, e^{i\alpha}, \sin\eta\, e^{i\beta}), \qquad 0 \le \eta \le \pi/2.$$

For $0 < \eta < \pi/2$, each fixed η gives a torus T_η. The sphere metric in these coordinates is

$$ds^2 = d\eta^2 + \cos^2\eta\, d\alpha^2 + \sin^2\eta\, d\beta^2.$$

At $\eta = 0$ the phase β ceases to describe a nonzero coordinate; at $\eta = \pi/2$ the analogous statement holds for α. The two endpoint sets are consequently circles, the cores of the solid tori from Chapter 2. The Clifford torus is the intermediate level $T = T_{\pi/4}$.

Define two actions of $\mathbb{R}/(2\pi\mathbb{Z})$ by

$$C_+(t; z) = (e^{it} z_1, e^{it} z_2), \qquad C_-(t; z) = (e^{it} z_1, e^{-it} z_2).$$

Both preserve η, hence every intermediate torus. Each action is free: a fixed state under a phase step requires $e^{it} = 1$, because at least one complex coordinate is nonzero. Every orbit has period 2π.

Proposition 3.1. *Each orbit of C_+ or C_- is a unit-speed great circle of S^3. On T, its coordinate winding is respectively $(1, 1)$ or $(1, -1)$.*

Proof. Set $v_+ = (iz_1, iz_2)$ and $v_- = (iz_1, -iz_2)$. In the real Euclidean inner product, each $v_\pm$ has norm one and is orthogonal to z. The corresponding orbit has the expression

$$C_\pm(t; z) = \cos t\, z + \sin t\, v_\pm.$$

It is therefore the unit circle in a real two-plane through the origin. Differentiating this expression gives speed one. The phase formulas increase (α, β) by (t, t) or $(t, -t)$, which gives the winding vectors after one full period. □

The coordinate curves with winding $(1, 0)$ and $(0, 1)$ also lie on T, but their radius in $\mathbb{R}^4$ is $1/\sqrt{2}$ and their affine planes do not pass through the origin. They are circles rather than ambient great circles. This distinction explains why the diagonal rulings, instead of the coordinate curves, are selected here. The standard Clifford-torus geometry is discussed in [11, §3.2]. Every individual orbit has the topology of S^1; the distinction lies in its embedding and its action convention.

3.2 Hopf observations and their fibers

The observation constant along a circle action records its orbit rather than the phase within it. For our two conventions the Hopf maps are

$$H_+(z) = (2z_1\overline{z_2}, |z_1|^2 - |z_2|^2), \qquad H_-(z) = (2z_1 z_2, |z_1|^2 - |z_2|^2). \tag{3.1}$$

Their values lie in $S^2 \subset \mathbb{C} \times \mathbb{R}$, since

$$4\,|z_1|^2\,|z_2|^2 + (|z_1|^2 - |z_2|^2)^2 = (|z_1|^2 + |z_2|^2)^2 = 1.$$

In angular coordinates their complex components are $\sin(2\eta)e^{i(\alpha-\beta)}$ and $\sin(2\eta)e^{i(\alpha+\beta)}$, with common real component $\cos(2\eta)$.

Proposition 3.2. *The fibers of H_+ are exactly the C_+ orbits, and the fibers of H_- are exactly the C_- orbits, including the core circles.*

Proof. Substitution shows that each map is constant on the stated orbits. A nonpolar value determines the component magnitudes and hence a unique $\eta \in (0, \pi/2)$. For H_+ it also determines $\alpha - \beta$ modulo 2π. Any two pairs of phases with that difference differ by a common phase, so they are on one C_+ orbit. For H_- the fixed sum gives opposite phase increments and one C_- orbit. At either polar value, one complex coordinate vanishes and the other has arbitrary unit phase. Its fiber is the corresponding core circle. □

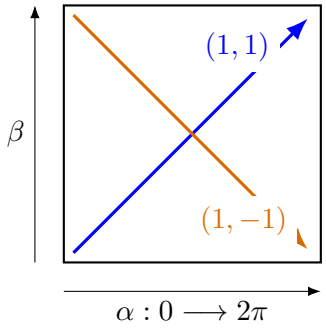
Thus a Hopf observation deliberately discards a continuous circle coordinate. Its ability to identify an orbit does not furnish a representative of the orbit. This simple example motivates the later distinction between a quotient reading and retained information sufficient to reconstruct a configuration. Lyons gives an elementary quaternion account of the Hopf construction [10]; here the phase formulas also prepare an explicit comparison of the two readings.

3.3 The paired reading and its antipodal ambiguity

Away from the two core circles, the pair $\mathcal{H} = (H_+, H_-)$ records η and the two phases

$$u = \alpha + \beta, \qquad v = \alpha - \beta \pmod{2\pi}.$$

The two base values are constrained to have the same latitude $\cos(2\eta)$. They are two compatible observations of one configuration, rather than independent arbitrary points of S^2.



Opposite sides are identified. The diagonals are great-circle rulings. Their determinant is -2: they span an index-two winding lattice.

Figure 3.1: An angular fundamental square for the Clifford torus. Opposite edges describe the same configurations, so the diagonal directions must be compared as integer windings as well as tangent directions.

Proposition 3.3. *For $z_1 z_2 \neq 0$, the fiber of $\mathcal{H}$ through z consists exactly of z and $-z$. A binary receipt recovers a representative after a phase-section convention has been fixed. At either core circle, the paired reading leaves a full phase undetermined.*

Proof. The phase transformation has matrix

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \det = -2.$$

A phase increment in its kernel satisfies $\alpha + \beta = \alpha - \beta = 0$ modulo 2π. Hence $\alpha = \beta$ modulo 2π and $2\alpha = 0$. The kernel is $\{(0,0), (\pi,\pi)\}$, giving precisely the two antipodal states.

For an explicit section, choose principal representatives $u, v \in [0, 2\pi)$ and set

$$z^0 = \left(\cos\eta\, e^{i(u+v)/2}, \sin\eta\, e^{i(u-v)/2}\right).$$

The retained sign $\epsilon \in \{+1, -1\}$ in $z = \epsilon z^0$ selects the state. At a core, both complex components of the paired output vanish; the remaining height is fixed and the nonzero state coordinate retains arbitrary phase. □

The principal-argument convention necessarily introduces cuts. To see their significance, keep η and v fixed and increase u by 2π continuously. A lifted state changes both α and β by π, ending at its antipode although the paired observation has returned to its initial value. Thus the two representatives cannot be distinguished by a single continuous section over the whole angular image. A retained binary choice supplies a representative, while continuity of a reconstruction is a local matter in a chosen phase neighborhood.

The determinant has a second interpretation in the winding lattice. Integer combinations of $(1,1)$ and $(1,-1)$ have coordinate windings of the same parity. Conversely, if (r,s) has that parity, then

$$(r,s) = \frac{r+s}{2}(1,1) + \frac{r-s}{2}(1,-1).$$

The two vectors span an index-two sublattice of $\mathbb{Z}^2$. They are a real basis of tangent directions, but they cannot replace an integer meridian–longitude basis without retaining the parity restriction. The phase ambiguity and the lattice index are therefore two expressions of the same integer matrix.

A third consequence is quantitative. Let the two S^2 outputs carry their Euclidean embedding norms and their product carry the sum-of-squares norm. Differentiating the angular formulas gives

$$\|D\mathcal{H}\ \mathrm{d}z\|^2 = 8\, d\eta^2 + 2\sin^2(2\eta)(d\alpha^2 + d\beta^2). \tag{3.2}$$

The cross terms from the sum and difference of phases cancel. Relative to the sphere metric, the squared singular values are

$$8, \qquad 8\sin^2\eta, \qquad 8\cos^2\eta.$$

Thus the paired map has injective differential away from the cores. On a region where $\sin\eta, \cos\eta \geq b > 0$, its local inverse differential, on the image tangent space and for a chosen representative, has norm at most $1/(\sqrt{8}\,b)$. This is a differential bound within a local branch. It does not remove the antipodal ambiguity or justify crossing an argument cut. At the cores the bound degenerates, in agreement with the invisible phase in Proposition 3.3.

The antipodal receipt should also be distinguished from the sheet receipt of orthogonal projection. The latter records the sign of a specified fourth coordinate. The former records which state lies above a paired phase value. They invert different observation maps, so agreement in the number of retained bits does not identify their contents.

3.4 Quaternion multiplication and the rotation group

Identify $\mathbb{R}^4$ with the quaternions by $q = z_1 + z_2 j$, with $ji = -ij$ and therefore $jz = \overline{z}\,j$ for $z \in \mathbb{C}$. Direct multiplication yields

$$e^{it}q = e^{it}z_1 + e^{it}z_2 j, \qquad qe^{it} = e^{it}z_1 + e^{-it}z_2 j.$$

Left multiplication realizes C_+ and right multiplication realizes C_-. Associativity gives their commutation. Multiplicativity of quaternion norm shows that both preserve the Euclidean state norm. In real coordinates their infinitesimal matrices are

$$\operatorname{diag}(J, J), \qquad \operatorname{diag}(J, -J), \qquad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

These are full four-dimensional operators, each rotating both coordinate planes. Their coordinate action is exactly the one already proved to generate great circles.

The larger representation containing these circle actions is

$$\rho(a, b)q = aqb^{-1}, \qquad a, b \in S^3, \qquad \operatorname{Spin}(4) \cong SU(2) \times SU(2).$$

Here the unit quaternions realize $SU(2)$. In this convention $C_+(t) = \rho(e^{it}, 1)$ and $C_-(t) = \rho(1, e^{-it})$. The representation is the usual double covering of $SO(4)$ [12, 4]. Its kernel can be verified directly. If $aqb^{-1} = q$ for every q, then $a = b$ by taking $q = 1$. This common quaternion must commute with every quaternion, hence is real; unit norm leaves $a = b = \pm 1$. Thus

$$\ker\rho = \{(1, 1), (-1, -1)\}.$$

This sign concerns two descriptions of one rotation. In contrast, the fiber $\{z, -z\}$ of the paired Hopf observation concerns two states giving one output. Their common twofold structure occurs in different spaces and acts on different objects.

Our Clifford convention is $\mathrm{Cl}_{0,2} \cong \mathbb{H}$. Two abstract generators anticommute and each has square -1; their product supplies the third imaginary quaternion. For four ambient coordinates the corresponding algebra is $\mathrm{Cl}_{0,4}$, with four anticommuting negative-square generators. Its even subalgebra is $\mathbb{H} \oplus \mathbb{H}$, providing the two quaternion factors associated with the spin representation [4]. The two-generator presentation describes a quaternion component; the four-generator algebra supplies the full ambient Clifford construction.

The quadratic form specifying the generator relations must be distinguished from the norm on configurations. The latter remains the positive Euclidean form $|z_1|^2 + |z_2|^2$. The negative generator squares specify an algebra convention; they do not themselves prescribe a physical time direction or a Lorentzian metric. Likewise, a Boolean sign schedule on a tape becomes an algebraic action only after a representation assigns its symbols to these operators. The representation is the data through which an operation acquires a geometric meaning.

There is an explicit isometry interchanging the selected families:

$$A(z_1, z_2) = (z_1, \overline{z_2}), \qquad AC_+(t)A^{-1} = C_-(t).$$

Complex conjugation reverses orientation in the second real plane, so A reverses orientation in $\mathbb{R}^4$. Choosing the two families therefore includes an orientation convention. A change of coordinates can exchange their roles while preserving the abstract circle topology. Retaining the coordinate convention makes the comparison unambiguous.

3.5 Finite clocks and exact intertwining

Choose one orbit and an integer $N \geq 3$. Its uniform samples are

$$e_z(k) = C_+(2\pi k/N; z), \qquad k \in \mathbb{Z}/N\mathbb{Z}.$$

The clock shift $\sigma_j(k) = k + j$ satisfies

$$e_z(\sigma_j(k)) = C_+(2\pi j/N; e_z(k)). \tag{3.3}$$

This exact identity is the comparison between the permutation and the continuous action. It records the embedding, the time step, and the state on which the operation acts.

For completeness, the comparison extends to linear representations. Let δ_k be the standard basis of $\mathbb{R}^N$, let $\Pi_j \delta_k = \delta_{k+j}$, and define $E_z \delta_k = e_z(k)$. Then

$$E_z \Pi_j = C_+(2\pi j/N) E_z.$$

The linear map E_z has a two-dimensional image, the orbit plane. Its restriction to $\{\delta_0, \ldots, \delta_{N-1}\}$ is injective, but its extension to all of $\mathbb{R}^N$ has a kernel of dimension $N - 2$. Thus exact agreement of the sampled operations does not imply an isomorphism between their entire ambient linear spaces.

The sampling error follows from the great-circle formula. For two phases,

$$\|C_+(t; z) - C_+(s; z)\| = 2\,|\sin((t - s)/2)|\,.$$

A nearest sample has phase error at most π/N, hence chord error at most $2\sin(\pi/(2N))$. The bound concerns one orbit; its base position under H_+ is still part of the configuration. An arbitrary permutation of clock labels may destroy cyclic adjacency. Interpreting a reordered tape therefore requires the permutation or its inverse as part of the representation.

3.6 The sixfold symmetry and its two sectors

For the polygonal construction, rotations through $\theta = \pi/3$ in either coordinate plane preserve the cells of K and F. Lemma 2.4 shows that g commutes with them, so Ψ intertwines the product

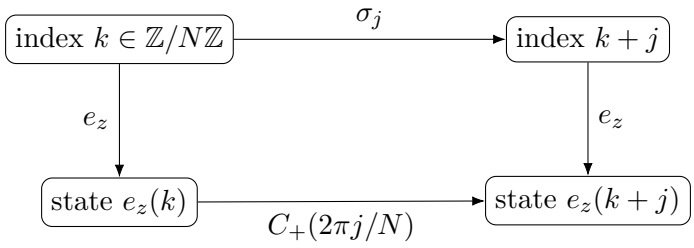


Figure 3.2: The clock permutation and the sampled continuous motion agree through a commuting diagram. The vertical maps specify which continuous states the clock labels represent.

symmetry $(\mathbb{Z}/6\mathbb{Z})^2$ with the corresponding round coordinate rotations. On the vertex labels the two selected great-circle steps are

$$S_+(j,k) = (j+1, k+1), \qquad S_-(j,k) = (j+1, k-1).$$

Each has order six, and they commute. Their generated translations have the form $(a+b, a-b)$ modulo six.

Proposition 3.4. *The group generated by S_+ and S_- has 18 elements. It has two orbits on the 36 vertex labels, distinguished by $j + k$ modulo two. The labels $(0,0)$ and $(1,0)$ may be chosen as orbit seeds.*

Proof. The homomorphism $(a,b) \mapsto (a+b, a-b)$ on $(\mathbb{Z}/6\mathbb{Z})^2$ has kernel $\{(0,0),(3,3)\}$. Its image has $36/2 = 18$ elements, and every image label has even coordinate sum. There are exactly 18 such labels, so this necessary parity condition also characterizes the image. Translations by that subgroup have the subgroup and its odd-sum coset as their two orbits. □

The same calculation explains when the phenomenon persists under resampling. On $(\mathbb{Z}/N\mathbb{Z})^2$, the kernel size is $\gcd(2, N)$. For odd N the two diagonal steps generate all translations. For even N they generate an index-two subgroup. The sixfold example is therefore the finite even-clock manifestation of the winding determinant, rather than an accidental omission in an orbit enumeration. A coordinate step $(j,k) \mapsto (j+1, k)$ exchanges the two sixfold sectors.

Finally, any round action can be transported to F by conjugation with Ψ. It is then a homeomorphism preserving the transported round metric. Preservation of the polyhedral cells requires the additional discrete symmetry proved above. Both great-circle families have finite permutation descriptions and quaternion descriptions; assigning one language to each expresses a choice of instrumental role. The exact intertwining maps state the content of their agreement. The topology of what remains visible, and the auxiliary information needed to recover what has been forgotten, are the subject of the next chapter.

Chapter 4

Observation, cohomology, and retained information

The sphere constructed in Chapter 2 is a complete state space, whereas an observation selects particular functions on that space. A projected interior, a visible loop, and a choice of lift belong to different levels of this construction. The interior belongs to an image; the loop belongs to a specified domain; the lift supplies information suppressed by a map. Their relationship becomes precise once the domain and the observation are stated together.

This chapter develops that relationship in two stages. First, it treats recovery as an inverse problem on observation fibers. Second, it constructs a finite complex on the separating torus and computes its cohomology with a specified inner product. The resulting harmonic coordinates provide the exact objects whose stability will be studied in Chapter 5. The sphere identification continues to rest on the explicit homeomorphism Ψ, rather than on a count of visible holes or measured periods.

4.1 Enclosure and accessible domains

The homology of the whole enclosure is

$$H_1(S^3;\mathbb{Z}) = 0, \qquad H_3(S^3;\mathbb{Z}) = \mathbb{Z}.$$

One computation uses a CW structure with one zero-cell and one three-cell. Its cellular differentials vanish, giving these groups directly. Cellular homology and its agreement with singular homology are developed in Hatcher's second chapter [6, §2.2]. On the polyhedral boundary F, the consistently oriented sum of its three-dimensional facets is a representative of the top-dimensional generator. Adjacent facet boundaries cancel, and there are no four-cells in F that could make this cycle a cellular boundary. The homeomorphism Ψ carries its orientation class to the fundamental class of the sphere.

The separating torus T has a different homology group: $H_1(T;\mathbb{Z}) = \mathbb{Z}^2$. This difference does not depend on an observational defect. It follows from changing the domain. The distinction is especially transparent when the two core circles

$$U = \{(e^{i\alpha}, 0)\}, \qquad V = \{(0, e^{i\beta})\}$$

are excluded. Their complement has global coordinates

$$S^3 \setminus (U \cup V) \cong T^2 \times (0, \pi/2), \qquad (\alpha, \beta, \eta) \longmapsto (\cos\eta\, e^{i\alpha}, \sin\eta\, e^{i\beta}).$$

For any fixed $\eta_0 \in (0, \pi/2)$, the homotopy replacing η by $(1-t)\eta + t\eta_0$ is a deformation retraction onto the torus at η_0. Both angular periods therefore survive throughout this accessible region.

A coordinate loop on T nevertheless contracts in the full sphere. Holding β fixed while increasing η to $\pi/2$ shrinks the α loop to $(0, e^{i\beta})$. Holding α fixed while decreasing η to zero similarly contracts a β loop. The endpoint of either contraction belongs to an excluded core, so neither homotopy is a contraction within the accessible region. This identifies exactly which domain restriction preserves the two periods.

Linking supplies a related distinction. Two distinct Hopf fibers can be linked although either fiber, considered alone, is null-homotopic in S^3. The obstruction to contracting one while avoiding the other lies in a complement. An explicit account of the linked fibers appears in [10]. A claim about a hole consequently needs its domain: the whole enclosure, its separating torus, a core complement, or a complement of selected fibers. These spaces can support different cycles while remaining parts of the same geometric construction.

4.2 Observation fibers and the recovery problem

Definition 4.1. An *exact observation* on a state domain D is a map $O : D \to Y$. Its fiber at y is $O^{-1}(y)$. A *retained lifting datum*, also called a receipt, is a map $R : D \to A$ whose value is retained together with the observation. It supports exact recovery when the combined map $(O, R) : D \to Y \times A$ is injective.

The word "receipt" will refer only to this retained mathematical information. It may be stored in a larger configuration, in a finite register, or in a record of operations. Those implementations can have different costs; the recovery question asks which states their values distinguish.

An observation determines the equivalence relation

$$s \sim_O t \quad \Longleftrightarrow \quad O(s) = O(t).$$

Let $\pi : D \to D/\!\sim_O$ be the quotient map. As sets, the quotient and $O(D)$ are in bijection through $[s] \mapsto O(s)$. Thus a quotient names the observable distinctions that remain after all states in a fiber have been identified.

Proposition 4.2 (Descent and recovery)**.** *A reader $q : D \to Z$ factors through π if and only if q is constant on each observation fiber. Exact recovery of every state from the pair $(O(s), R(s))$ is possible if and only if R is injective on each fiber of O.*

Proof. If $q = \bar{q} \circ \pi$, equal classes have equal readings. Conversely, constancy on each class makes $\bar{q}([s]) = q(s)$ well-defined, and this is the required factorization. For the second assertion, injectivity on every fiber is equivalent to injectivity of (O, R). An injective map has an inverse on its image, while any proposed recovery map would force injectivity by recovering equal outputs to equal states. □

This proposition is algebraic. Continuity and a quantitative inverse estimate require additional hypotheses. For example, a continuous injective map from a compact state space to a Hausdorff output space is a homeomorphism onto its image. A discontinuously chosen lifting label need not satisfy that hypothesis. In the orthogonal projection, the sheet choice separates the two lifts away from the silhouette, while the two lifts coincide at the silhouette itself. A stable continuous inverse is therefore stated on a fixed sheet with a positive margin, as in Proposition 2.6.

Observation	Identified information	Retained lifting datum
Orthogonal P	Two fourth-coordinate signs	Sheet, except at the silhouette
Paired H_+, H_-	z and $-z$ off the cores	Antipodal bit; full phase at a core
Spin action ρ	(a, b) and $(-a, -b)$	Lift of a rotation to the acting group
Scalar equation $r(y) = 0$	Choice among solution roots	Selected interval or root branch
Constant quotient	Every admitted state	A retained representative or its code

Table 4.1: Different observation maps give different lifting problems. Equal fiber cardinalities alone do not identify their state spaces or inverses.

The paired Hopf reading illustrates why a discrete ambiguity can persist even when local inverse formulas exist. Away from the cores, the observable angles are $u = \alpha + \beta$ and $v = \alpha - \beta$ modulo 2π. The corresponding torus map has integer matrix

$$B = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \det B = -2.$$

Its fibers have two points. There is no continuous global choice between them. Indeed, a continuous section would induce an integer matrix C on the fundamental groups of the tori satisfying $BC = I$. Taking determinants would give $-2 \det C = 1$, which is impossible over $\mathbb{Z}$. Local choices can be made on angle charts, but crossing their cuts requires the retained choice described in Chapter 3. At a core, an entire phase disappears, so a binary label no longer suffices.

The other rows of Table 4.1 involve independent choices. For example, $13y^2 - 36y + 18 = 0$ has roots $(18 \pm 3\sqrt{10})/13$. Selecting the lower root fixes a scalar branch. It specifies no sign of w in (2.2) until an encoding connects the scalar variable with that geometric coordinate. Likewise, the kernel of the spin action concerns a representation of an acting group, rather than the antipodal ambiguity of a state observation.

A quotient generated by a total order gives a more extreme example. If every comparable pair is identified, then every pair of elements is identified, because a total order makes every pair comparable. The resulting quotient has one point, and every descended numerical reader is constant. A larger output may retain an integer alongside that quotient. Recovery then uses this retained integer; it cannot be attributed to distinctions within the one-point quotient.

4.3 Finite ambiguity and approximate readings

Exact fibers should be distinguished from numerical neighborhoods. For $\varepsilon > 0$, the relation $\|O(s) - O(t)\| \le \varepsilon$ need not be transitive: the real observations $0, \varepsilon, 2\varepsilon$ already provide a counterexample. An error ball describes the states compatible with a bounded discrepancy. It becomes an equivalence class only after some further construction, whose consequences must then be examined separately.

There is an elementary information requirement for exact recovery on finite fibers.

Proposition 4.3 (Size of retained information)**.** *If an observation fiber contains m admitted states, any receipt alphabet that supports exact recovery on that fiber has at least m symbols. A fixed-length binary encoding requires at least $\lceil \log_2 m \rceil$ bits in that worst case.*

Proof. An alphabet with fewer than m symbols assigns the same receipt to two states in the fiber. Their observations are already equal, so the combined observation and receipt cannot be injective. A binary word of length b has 2^b possible values; hence $2^b \ge m$. □

For a uniform random state S on a finite set of size N, let m_y be the size of the fiber at observation y. The conditional distribution on that fiber is uniform. Its conditional entropy, measured in bits, is therefore

$$H(S \mid O(S)) = \sum_{y \in O(D)} \frac{m_y}{N} \log_2 m_y.$$

For the 36-state construction of Chapter 6, the orthogonal reader has twelve singleton fibers and twelve double fibers. Its conditional entropy is $24/36 = 2/3$ bit. The paired Hopf reader has eighteen double fibers and conditional entropy one bit. A constant reader leaves $\log_2 36$ bits. The worst-case alphabet bound and this average information quantity answer different questions, and the latter depends on the stated input distribution. Neither calculation by itself assigns an energy cost to recording or erasing information.

4.4 A cellular model of the accessible torus

Take a unit square with opposite sides identified, and divide it by the diagonal from $(0,0)$ to $(1,1)$. The quotient has one vertex, oriented horizontal and vertical edges a, b, a diagonal edge c, and two triangular two-cells f_1, f_2. This is a CW description: the edges become loops, and the face boundaries are attaching maps rather than boundaries of embedded Euclidean triangles after identification.

The oriented attaching words are abc^{-1} and $ca^{-1}b^{-1}$. Passing to cellular chains over $\mathbb{R}$ gives

$$\partial a = \partial b = \partial c = 0, \qquad \partial f_1 = a + b - c, \qquad \partial f_2 = -a - b + c.$$

The two face boundaries are negatives. Their sum is the fundamental two-cycle of the torus. The image of ∂_2 is one-dimensional, so

$$H_1(T;\mathbb{R}) = \mathbb{R}^3 / \operatorname{span}\{a + b - c\}, \qquad H_2(T;\mathbb{R}) = \operatorname{span}\{f_1 + f_2\}.$$

In particular, the diagonal represents $a + b$ in first homology. The cell structure has torus topology because it was obtained from the identified square, not because its alternating cell count happens to vanish.

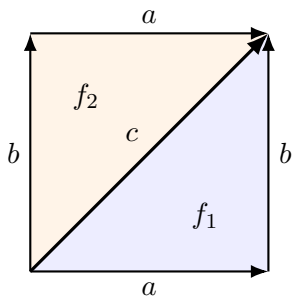
Choose the cell cochains dual to these oriented cells and equip their coordinate spaces with standard Euclidean inner products. The cochain complex is

$$C^0 = \mathbb{R} \xrightarrow{d_0 = 0} C^1 = \mathbb{R}^3 \xrightarrow{d_1 = \begin{pmatrix} 1 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix}} C^2 = \mathbb{R}^2. \tag{4.1}$$

A cochain $\alpha = (x, y, z)$ assigns the three periods x, y, z to the oriented loops. Closedness is exactly $x + y - z = 0$. Since $d_0 = 0$, no nonzero one-cochain is exact, and every cohomology class has one representative in $\ker d_1$. Thus

$$H^1(T;\mathbb{R}) = \{(A, B, A + B) : A, B \in \mathbb{R}\}.$$

The integral periods form the lattice obtained by taking $A, B \in \mathbb{Z}$. The diagonal contributes a consistency relation between the two independent periods rather than a third independent period.



Measured domain: T^2
$\dim \mathcal{H}^1 = 2$
Whole enclosure: S^3
$H_1(S^3) = 0$, $H_3(S^3) = \mathbb{Z}$

Figure 4.1: Opposite edges and all four corners are identified. The accessible torus has its own period domain within the whole enclosure.

4.5 Harmonic representatives and measured periods

For a finite cochain complex with inner products, the degree-one Hodge operator is $\Delta_1 = d_0 d_0^* + d_1^* d_1$. Its kernel consists of closed cochains orthogonal to exact cochains. Indeed,

$$\langle \Delta_1 \alpha, \alpha \rangle = \|d_0^* \alpha\|^2 + \|d_1 \alpha\|^2 .$$

Moreover $\ker d_1 = \operatorname{im} d_0 \oplus \mathcal{H}^1$ orthogonally: project a closed cochain onto $\operatorname{im} d_0$, and the remaining component is still closed and annihilated by d_0^*. Consequently every cohomology class has a unique harmonic representative. This finite-dimensional form of Hodge theory underlies combinatorial inference [7].

For the selected torus, write $v = (1, 1, -1)^T$. Direct multiplication gives

$$\Delta_1 = 2vv^T, \qquad \mathcal{H}^1 = \{(A, B, A + B) : A, B \in \mathbb{R}\}. \tag{4.2}$$

The operator vanishes on the two-dimensional plane perpendicular to v and sends v to $6v$. Its only positive eigenvalue is therefore 6. These numerical values use the Euclidean cell inner products already specified; the dimension of cohomology follows from incidence, whereas the positive spectral value also uses the metric.

The period map $M(x, y, z) = (x, y)$ reads a, b. Its restriction to $\mathcal{H}^1$ is bijective onto $\mathbb{R}^2$, with inverse

$$(A, B) \longmapsto (A, B, A + B).$$

The inequality

$$\|(A, B, A + B)\|^2 = A^2 + B^2 + (A + B)^2 \leq 3(A^2 + B^2)$$

shows that this inverse has norm at most $\sqrt{3}$. Equality at $A = B$ proves that the norm is exactly $\sqrt{3}$, or equivalently that the smallest harmonic response is $\gamma = 1/\sqrt{3}$. One period cannot be injective on a two-dimensional harmonic space. Two periods determine the existing harmonic representative without changing the complex that supports it.

Removing both faces while retaining the three loop edges makes d_1 vanish and increases the harmonic dimension to three. Attaching a disk along a instead makes that cycle a boundary. Both operations alter the cellular model. Reducing a measured circulation on an unchanged complex performs neither operation: a small cochain coefficient is information about a cochain, not an attachment of a new cell.

4.6 Transport of representation and metric

A change of coordinates preserves the preceding calculations when it carries both incidence and inner product. Let $S_k : C^k \to C'^k$ be invertible maps satisfying

$$d'_k S_k = S_{k+1} d_k, \qquad \langle S_k u, S_k v \rangle' = \langle u, v \rangle .$$

The second condition makes S_k an isometry for the transported inner products. Taking adjoints of the first gives

$$(d'_k)^* S_{k+1} = S_k d_k^*, \qquad \Delta'_1 S_1 = S_1 \Delta_1.$$

Hence harmonic representatives and the complete spectrum are transported together. If the period reader is transported as $M' = M S_1^{-1}$ and its output norm is fixed, its smallest harmonic response is also unchanged.

An incidence-preserving isomorphism without metric transport still identifies the cohomology groups. It need not preserve adjoints, harmonic representatives as coordinate vectors, or positive eigenvalues. Arbitrarily changing cell weights is a change in the calibrated geometry of the finite problem. Compatible maps and controlled inner products play a central role in finite element exterior calculus [1, §3]. Here their finite-dimensional form separates the topological statement that two periods exist from the quantitative statement that those periods can be recovered with a particular constant.

Chapter 5

Stability of observation and reconstruction

Exact recovery asks whether an output determines a state. Stable recovery asks how much that state can change when the output changes. The first question is settled by fibers, retained lifting data, and the incidence structure of the accessible complex. The second requires norms and positive quantitative margins. An injective map can have a poorly conditioned inverse; an accurate observation can also leave an entire fiber unresolved.

The present chapter develops stability after the exact choices have been fixed. Geometry supplies the inverse bound on a selected projection sheet. The cochain complex supplies a residual estimate transverse to its harmonic space, and measured periods supply control within that space. A retained word of algebraic operations supplies a further bound for accumulated computational error. These estimates meet in a comparison of two representations of one calibrated observable. Their constants come from the respective operators and recovery domains.

5.1 Residual energy and the harmonic component

Let

$$C^0 \xrightarrow{d_0} C^1 \xrightarrow{d_1} C^2, \qquad d_1 d_0 = 0,$$

be a finite real cochain complex. Equip each space with a positive definite inner product, and form the corresponding adjoints. Throughout this section all norms are induced by those fixed inner products. Define

$$\Delta_1 = d_0 d_0^* + d_1^* d_1, \qquad \mathcal{H}^1 = \ker \Delta_1.$$

The energy identity

$$\langle \Delta_1 \alpha, \alpha \rangle = \|d_0^* \alpha\|^2 + \|d_1 \alpha\|^2$$

shows that $\mathcal{H}^1 = \ker d_0^* \cap \ker d_1$. The first residual measures failure to be orthogonal to exact cochains, and the second measures failure of closedness. Neither residual detects a harmonic component. Finite Hodge theory gives the appropriate orthogonal separation [7, 1].

Write $\alpha = h + e$, where h is the orthogonal projection of α onto $\mathcal{H}^1$ and $e \perp \mathcal{H}^1$. When the orthogonal complement is nonzero, its smallest eigenvalue under Δ_1 is positive. Denote this eigenvalue by λ. If the complement is zero, every cochain is harmonic and the transverse estimate is unnecessary.

Proposition 5.1 (Residual estimate)**.** *Suppose $\mathcal{H}^{1\perp} \neq \{0\}$ and the least eigenvalue of Δ_1 on this space is $\lambda > 0$. Then*

$$\|e\| \leq \lambda^{-1/2} \left(\|d_0^* \alpha\|^2 + \|d_1 \alpha\|^2 \right)^{1/2}.$$

Equality is possible when e belongs to the eigenspace of λ.

Proof. Expand e in an orthonormal eigenbasis of the self-adjoint operator Δ_1. Every eigenvalue on the orthogonal complement is at least λ, so $\lambda \|e\|^2 \leq \langle \Delta_1 e, e \rangle$. The harmonic component contributes zero to both residuals. The energy identity then gives the estimate, and an eigenvector at λ gives equality. □

The estimate controls distance to the harmonic space. It cannot determine which harmonic representative is present. In particular, adding any $h_0 \in \mathcal{H}^1$ leaves both residuals unchanged. This exact kernel is the reason that further period measurements are required; decreasing a residual tolerance does not remove it.

5.2 Period measurements and uniqueness

Let $M : C^1 \to \mathbb{R}^p$ be a linear period reader, with a specified Euclidean norm on its output. Suppose $\mathcal{H}^1$ is nonzero. Define the smallest harmonic response by

$$\gamma = \min_{h \in \mathcal{H}^1,\, \|h\|=1} \|Mh\|.$$

Compactness of the unit sphere shows that the minimum is attained. It is positive exactly when $M|_{\mathcal{H}^1}$ is injective. Thus γ expresses the quantitative strength of a qualitative uniqueness assertion.

Proposition 5.2 (Recovery from periods)**.** *Assume $\gamma > 0$. For data $m = Mh + \xi$ with $h \in \mathcal{H}^1$ and $\|\xi\| \leq \nu$, the unique least-squares estimate on $\mathcal{H}^1$ satisfies*

$$\left\| \widehat{h} - h \right\| \leq \nu / \gamma. \tag{5.1}$$

If the same measurement protocol instead supplies $m = M\alpha + \xi$ for $\alpha = h + e$, the estimate of its harmonic component obeys

$$\left\| \widehat{h} - h \right\| \leq \frac{\nu + \|M\| \, \|e\|}{\gamma}.$$

Proof. Choose orthonormal coordinates on $\mathcal{H}^1$. The restriction $B = M|_{\mathcal{H}^1}$ has full column rank and least singular value γ. Its least-squares inverse is its Moore–Penrose pseudoinverse $B^\dagger$, with norm $1/\gamma$. Hence $\widehat{h} - h = B^\dagger \xi$. In the second protocol, the effective data error is $Me + \xi$; apply the triangle inequality before the same inverse bound. □

The two protocols correspond to different orders of operation. One may first remove the transverse component and then measure periods, or measure the uncorrected cochain and account for its contribution to the data. The stated inverse estimate identifies which protocol is being analyzed. Singular-value methods provide the general linear-algebraic setting for these statements [5, Chapter 5].

The residual and the period reader can also be treated as a single observation. Give the output below the product inner product and define

$$\mathcal{A}\alpha = (d_0^* \alpha, d_1 \alpha, M\alpha).$$

Theorem 5.3 (Complete linear observation)**.** *The operator $\mathcal{A}$ is injective if and only if M is injective on $\mathcal{H}^1$. When this condition holds and $C^1 \neq \{0\}$, its least singular value τ is positive. For any two cochains,*

$$\|\alpha - \beta\| \leq \frac{1}{\tau} \|\mathcal{A}\alpha - \mathcal{A}\beta\| .$$

Least-squares reconstruction from data with error at most $\nu_{\mathcal{A}}$ has state error at most $\nu_{\mathcal{A}}/\tau$.

Proof. The first two components of $\mathcal{A}\alpha = 0$ imply $\alpha \in \mathcal{H}^1$. Its remaining component then vanishes only at zero precisely when $M|_{\mathcal{H}^1}$ is injective. In finite dimensions, injectivity gives a positive least singular value. Apply its defining inequality to $\alpha - \beta$, or apply the corresponding pseudoinverse to the data error. □

A numerical value for τ presupposes the output norm, including any weights between residuals and periods. Rescaling one output component changes the singular values of the combined observation without changing its kernel. Such scaling is a calibration choice and belongs in the mathematical description of the reader.

5.3 The torus calculation in full

For the torus of (4.1), $d_0 = 0$, $\lambda = 6$, and $\gamma = 1/\sqrt{3}$. If $\alpha = (x, y, z)$ and $\rho = x + y - z$, the orthogonal decomposition is

$$e = \frac{\rho}{3}(1, 1, -1), \qquad h = \alpha - e, \qquad \|e\| = \frac{|\rho|}{\sqrt{3}} = \frac{\|d_1\alpha\|}{\sqrt{6}}.$$

Thus every transverse component lies in the sole positive eigenspace, and the residual bound is always exact. Corrected period data (m_a, m_b) recover

$$\widehat{h} = (m_a, m_b, m_a + m_b).$$

The reconstruction operator has norm $\sqrt{3}$, as calculated in the preceding chapter. Errors with equal components in the two periods attain this amplification.

The combined observation also has an exact singular-value calculation. With the Euclidean product norm and $M(x, y, z) = (x, y)$,

$$\mathcal{A}^*\mathcal{A} = \begin{pmatrix} 3 & 2 & -2 \\ 2 & 3 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

The vector $(1, -1, 0)$ is an eigenvector with eigenvalue 1. On the span of $(1, 1, 0)/\sqrt{2}$ and $(0, 0, 1)$, the matrix is

$$\begin{pmatrix} 5 & -2\sqrt{2} \\ -2\sqrt{2} & 2 \end{pmatrix},$$

whose characteristic polynomial is $t^2 - 7t + 2$. Consequently

$$\tau = \sqrt{\frac{7 - \sqrt{41}}{2}} > 0.$$

This gives a direct inverse bound for the complete cochain from its two face residuals and two measured periods. The separate constants λ and γ explain the two sources of control, while τ packages them into a single specified data norm.

5.4 Perturbations and calibration margins

A positive inverse margin can survive a bounded change of the reader. Let $\widetilde{M}$ be another period map on the same spaces, and suppose $\left\|\widetilde{M} - M\right\| \le \zeta < \gamma$. Then, for every $h \in \mathcal{H}^1$,

$$\left\|\widetilde{M}h\right\| \ge \|Mh\| - \left\|(\widetilde{M} - M)h\right\| \ge (\gamma - \zeta)\,\|h\|\,.$$

Thus the perturbed restriction remains injective, and its inverse norm is at most $(\gamma - \zeta)^{-1}$. The assumption fixes the harmonic space and both norms; it compares operators on the same calibrated problem.

If data were generated by M but reconstruction uses $\widetilde{M}$, the calibration discrepancy contributes a bias. For $m = Mh + \xi$, least-squares reconstruction using $\widetilde{M}|_{\mathcal{H}^1}$ satisfies

$$\left\|\widehat{h} - h\right\| \le \frac{\nu + \zeta\,\|h\|}{\gamma - \zeta}.$$

A uniform numerical bias allowance therefore also needs a bound on the admitted state amplitudes. Merely proving that the perturbed reader is invertible does not supply that bound.

For a self-adjoint positive semidefinite operator $\widetilde{\Delta}_1$ on the same inner-product space, assume that

$$\ker \widetilde{\Delta}_1 = \mathcal{H}^1, \qquad \left\|\widetilde{\Delta}_1 - \Delta_1\right\| \le \beta < \lambda.$$

On the common orthogonal complement, the Rayleigh quotient gives

$$\left\langle \widetilde{\Delta}_1 e, e \right\rangle \ge (\lambda - \beta)\,\|e\|^2\,.$$

Its gap is at least $\lambda - \beta$. The shared-kernel hypothesis is essential to this particular estimate. A different attachment map can alter the harmonic space, while a different cell metric can change which coordinate vectors represent it. Such changes first require an identification of spaces and metrics before an operator perturbation can be interpreted.

For the explicit torus, a period-map perturbation of at most $1/100$ leaves response at least $1/\sqrt{3} - 1/100 > 0$. A same-kernel Hodge-operator perturbation of at most $1/10$ leaves a gap of at least $59/10$. These are conservative margins, expressed in the specified operator norms. They illustrate robustness of the fixed inverse problem and do not treat a new cell attachment as a calibration error.

5.5 Composition of operations and the role of order

Let an exact word of operations be $T_n \circ \cdots \circ T_1$. Suppose T_j is L_j-Lipschitz on a domain containing both the exact and computed inputs. Let the computed step have error at most ε_j relative to T_j applied to that computed input. If E_j is the state error after the jth step, then

$$E_j \le L_j E_{j-1} + \varepsilon_j.$$

Successive substitution proves

$$E_n \le E_0 \prod_{j=1}^{n} L_j + \sum_{j=1}^{n} \varepsilon_j \prod_{k=j+1}^{n} L_k. \tag{5.2}$$

The products describe how later operations amplify an earlier error. Round quaternion actions are isometries for the chord metric, so their L_j equal one and the bound reduces to addition of step

errors. Quantization can be included as a bounded step error when its rule and norm are stated. A different generator or a lost inverse permutation changes the exact word being compared and needs a corresponding change in the observation model.

Order can be studied by comparing two composites on the same state domain. An example inherited from the instrument has a complete calculation. For unit vectors e, f in a real inner-product space, put

$$Q_e = I - ee^T, \qquad Q_f = I - ff^T, \qquad c = \langle e, f \rangle .$$

Multiplication gives

$$[Q_e, Q_f] = c(ef^T - fe^T), \qquad \|[Q_e, Q_f]\| = |c|\sqrt{1-c^2} \le \tfrac{1}{2}.$$

For $|c| < 1$, choose $g = (f - ce)/\sqrt{1-c^2}$. On the orthonormal basis e, g, the commutator has matrix

$$\begin{pmatrix} 0 & c\sqrt{1-c^2} \\ -c\sqrt{1-c^2} & 0 \end{pmatrix};$$

it vanishes on the orthogonal complement. The endpoint cases $|c| = 1$ give zero directly. The inequality follows from $c^2(1-c^2) \le 1/4$.

Its exponential is an invertible rotation supported on this one plane, with angular speed $|c|\sqrt{1-c^2}$. A Clifford bivector represents that planar rotation. The projections themselves are noninvertible. Moreover, the commutator has rank at most two, whereas the generators $\mathrm{diag}(J, \pm J)$ of nontrivial Hopf motion have rank four. The exponential therefore defines a precise local rotation operation whose relationship to a Hopf action requires a comparison map. Exponentiating an order defect does not by itself identify the two actions.

5.6 Comparison through a common observable

The preceding estimates concern different spaces. They can be compared by specifying the observable they are intended to recover. Let $q : D \to Y$ be a dimensionless calibrated observable in a normed space. For representation i, let $E_i : D \to Z_i$ be an encoding and $r_i : Z_i \to Y$ its reader. Suppose

$$\|r_i(E_i(s)) - q(s)\| \le \eta_i, \qquad \|\widehat{z}_i - E_i(s)\| \le \delta_i,$$

where η_i bounds the exact diagram defect and δ_i bounds the state error. Assume that r_i is L_i-Lipschitz throughout the relevant recovery domain.

Proposition 5.4 (Comparison of representations)**.** *Under these assumptions, observations of the same admitted state satisfy*

$$\|r_1(\widehat{z}_1) - r_2(\widehat{z}_2)\| \le \eta_1 + \eta_2 + L_1\delta_1 + L_2\delta_2. \tag{5.3}$$

Proof. Insert $r_1(E_1(s))$, then $q(s)$, then $r_2(E_2(s))$ between the two computed readings. The triangle inequality bounds the four successive differences, and the reader Lipschitz constants bound the two differences arising from state error. □

The comparison requires one specified q on one admitted state domain. Similar formulas in different representations do not establish this common observable by themselves. The diagram defects record the discrepancy remaining after the encodings and readers have been chosen. Conversely, agreement of two computed readings supplies no inverse statement unless the common observation, together with its retained data, distinguishes the states in question.

Structure	Exact data fixed	Stability constant or bound
Geometry	Sheet and inverse domain	$\delta x/a$ for lifted state error
Algebra	Generator word and action map	Equation (5.2)
Topology	Incidences and period domain	$\lambda^{-1/2}$ residual, γ^{-1} period error
Common reading	Calibrated observable and readers	Equation (5.3)

Table 5.1: The inverse estimates apply after the discrete lifting choices and the exact domains have been specified.

5.7 Trace errors as inputs to reconstruction

A trace observable supplies one useful example of a bound that can enter this comparison. Under the earlier instrument's uniform-grid sampling, boundary, knot-noise, and Hermite-slope hypotheses, its estimate is

$$\|u - f\|_\infty \le \frac{7}{4}\Omega_f(h) + \frac{5}{2}\epsilon + \frac{h^2}{4\sqrt{6}}\|r\|_2, \qquad \Omega_f(h) \ge \omega_f(h).$$

Here h is the grid spacing, ϵ bounds knot errors, $\|r\|_2$ is the Euclidean norm of the slope residual, and ω_f is the modulus of continuity. The first term controls unresolved variation of the target, the second controls the admitted knot perturbations, and the last controls the error left by the slope calculation. The precise construction occurs in `instrument/chapters/05.tex`, in the passage headed "complete error budget." Classical minimum properties of splines supply the variational context [3, 2].

Using that estimate in a sphere-valued reconstruction requires the coordinate reader and its conversion to the sphere metric. A scalar trace error may be amplified by a subsequent inverse, especially near a projection silhouette. Its contribution then enters through the appropriate reader constant in (5.3). Coordinatewise accuracy does not establish a cell attachment, select an omitted sheet, or determine an unmeasured harmonic period.

All bounds in this chapter are deterministic. Statistical uncertainty requires an accompanying probability model and a stated coverage interpretation. A norm inequality alone supplies no confidence level; the distinction between a measurement model, uncertainty propagation, and coverage is treated in the metrological references [8, 9].

The limits of the inverse estimates can be seen in exact examples. The two points (x, w) and $(x, -w)$ have identical orthogonal observations but are separated by chord distance $2\,|w|$. Removing both certifying faces of the torus complex introduces an additional harmonic direction with zero local residual. These failures occur before any small numerical tolerance is considered. Once the domain, incidence structure, and retained lifting information are fixed, the positive constants proved above quantify reconstruction within that mathematical model.

Chapter 6

A finite realization and its reconstruction

The continuous construction admits a finite realization in which every state, observation fiber, and recovery map can be calculated. This realization supplies a useful meeting point for the preceding chapters. The product polytope provides its vertices, the two circle actions provide its operations, and the torus complex provides a two-period record. Quantitative reconstruction then depends on three explicit features: separation of the finite observations, a retained choice within each observation fiber, and the response of the period map.

We shall use two six-position clocks. This choice makes the connection with the hexagonal factors exact and keeps the observation fibers small enough to display in full. The resulting model is a finite sector of the continuous enclosure. Its calculations establish correspondence among specified mathematical descriptions of that sector; the question of interpreting an entire physical apparatus requires additional identifications. The finite example is useful precisely because its hypotheses and its conclusion can both be stated without such an identification.

6.1 States, encodings, and the action of the clocks

Let

$$D = (\mathbb{Z}/6\mathbb{Z})^2, \qquad \theta = \pi/3,$$

and choose representatives $0 \le j, k < 6$ whenever an integer is printed. Recall the vertices $u_j = (\cos j\theta, \sin j\theta)$ of the hexagon. The two geometric encodings are

$$E_F(j,k) = (u_j, u_k) \in F, \qquad E(j,k) = \Psi(E_F(j,k)) = \frac{1}{\sqrt{2}}(e^{ij\theta}, e^{ik\theta}). \tag{6.1}$$

At a polygon vertex the radial map of Chapter 2 reaches the unit circle. Consequently the displayed formula follows directly from the definition of Ψ. All 36 states lie on the separating Clifford torus, and their preimages are vertices of the product polytope.

Both encodings are injective. A sixth root of unity uniquely determines its exponent modulo six, so reading the two complex coordinates gives an inverse on $E(D)$. An alternative encoding uses the integer

$$n = j + 6k \in \{0, \dots, 35\}.$$

Division with remainder gives $j = n \bmod 6$ and $k = \lfloor n/6 \rfloor$. These are different codomains for the same admitted states, with explicit maps between their images. Their equivalence concerns the

retained information; it does not require the ambient integer interval to possess the geometry of the torus.

Let T_1 increment the first index and T_2 the second. Their actions on (6.1) multiply the corresponding complex coordinate by $e^{i\theta}$. The selected simultaneous steps are

$$S_+ = T_1 T_2, \qquad S_- = T_1 T_2^{-1}.$$

They therefore agree exactly with the parameter-θ restrictions of the two round circle actions of Chapter 3. This agreement is an intertwining identity between a permutation action and a geometric action, rather than a resemblance between their drawings.

The selected steps preserve $j + k$ modulo two. More explicitly, applying $S_+^a S_-^b$ changes the two indices by

$$(a + b, a - b) \pmod 6.$$

The kernel of this homomorphism from $(\mathbb{Z}/6\mathbb{Z})^2$ consists of $(0, 0)$ and $(3, 3)$. Its image has 18 elements and lies in the 18-element set of increments with even coordinate sum, hence equals that set. There are exactly two orbits: seed $(0, 0)$ generates the even sector, and seed $(1, 0)$ generates the odd sector.

A seed and a retained operation word thus determine the resulting state. The converse requires care. Commutativity, the sixth-power relations, and the displayed kernel allow several words to reach the same state. Recovering a final state recovers neither the ordering of its operations nor the number of complete turns. An observation of history must retain the word or an appropriately defined accumulated record. This distinction persists even when all state calculations are exact.

6.2 Orthogonal observation and the retained sheet

Choose the real coordinate order in which the orthogonal observation and the omitted coordinate are

$$x = \frac{1}{\sqrt{2}}(\cos j\theta, \sin j\theta, \cos k\theta), \qquad w = \frac{\sin k\theta}{\sqrt{2}}. \tag{6.2}$$

The first two observed coordinates recover j exactly on the finite image. The last gives $\cos k\theta$, and hence determines k up to $k \leftrightarrow -k$. For $k = 0, 3$ there is only one possibility; for $k = 1, 2, 4, 5$ there are two. Retain the sign of $\sin k\theta$ on these double fibers, using the canonical symbol 0 on the singleton fibers. The sheet lift (2.2) followed by the inverse of E then recovers the state.

For each fixed j, the two singleton fibers are $\{0\}$ and $\{3\}$, while the double fibers are $\{1, 5\}$ and $\{2, 4\}$. Orthogonal observation therefore has 24 distinct values: twelve singleton fibers and twelve double fibers. The sheet is additional data exactly where the observation has identified two states. Its necessity can be seen before introducing any numerical error: two members of a double fiber have identical observations.

The 24 states with $k \in \{1, 2, 4, 5\}$ form the stable subdomain

$$D_* = \{(j, k) \in D : k \in \{1, 2, 4, 5\}\}.$$

Here $|w| = \sqrt{3/8}$, so the continuous lift has a positive conditioning margin. The other twelve states lie on the silhouette $w = 0$. They remain distinct admitted states with unique exact observations. The loss of a uniform continuous inverse bound at the silhouette does not erase their finite codes; it concerns perturbations in the larger continuous observation domain. We shall see below that the finite observation alphabet still has positive separation there.

6.3 Paired circle observations and their antipodal fiber

The paired Hopf maps give a second observation. On $E(D)$ their third coordinates vanish, and their equatorial phases are

$$u = j + k \pmod 6, \qquad v = j - k \pmod 6.$$

We use $(e^{iu\theta}, e^{iv\theta}) \in \mathbb{C}^2$ as the numerical representation of this observation. Its image consists exactly of the 18 pairs with $u \equiv v \pmod 2$. To prove the assertion and construct the inverse, take the canonical representatives of such a pair and set

$$j_0 = \tfrac{1}{2}((u+v) \bmod 6) \in \{0,1,2\}, \qquad k_0 = (u - j_0) \bmod 6. \tag{6.3}$$

The parity condition makes the division by two an ordinary integer division. We have $j_0 + k_0 \equiv u$ and $j_0 - k_0 \equiv v$ modulo six, so this construction reaches every allowed observation. The kernel calculation from the preceding section shows that its two preimages are

$$(j_0, k_0), \qquad (j_0 + 3, k_0 + 3) \pmod 6.$$

Their geometric encodings are antipodal. A retained bit $b \in \{0,1\}$ selects the state $(j_0 + 3b, k_0 + 3b)$.

The bit has a different role from the parity of the orbit. The observation already gives $j + k \equiv u \pmod 2$, so it identifies the even or odd sector. Its remaining ambiguity is the antipodal pair inside that sector. Likewise, the antipodal bit belongs to the state observation; it is distinct from a choice of lift of a rotation through the spin covering. These distinctions follow from the domains of the maps, even when each ambiguity can be represented by two symbols.

Exact reader on D	Image size	Largest fiber	Retained recovery data
Full code $j + 6k$	36	1	None
Orthogonal $P \circ E$	24	2	Sheet on double fibers
Paired Hopf phases	18	2	Antipodal bit
Constant quotient	1	36	A representative code

Table 6.1: Observation fibers of the finite realization. The retained data complete each reader to an injective observation.

For the last row, take the quotient generated by the relation $n \le n'$ on the 36 integer codes. Totality identifies every pair in the generated equivalence relation, so the quotient has one element. Every numerical function descended to it is constant. The augmented output $(*, n)$, however, retains an integer code and has an exact inverse. This provides an extreme example of the distinction between an observation and its accompanying recovery data. The useful integer belongs to the larger output and cannot be reconstructed from the collapsed quotient alone.

6.4 Separation and finite decoding

For an injective finite encoding $A \to Y$ into a metric space, let $s > 0$ be the smallest distance between distinct image points. An observation at distance less than $s/2$ from the true image point has that point as its unique nearest image: the triangle inequality places every competing point farther away. Strict inequality matters, since an observation on a midpoint can have two nearest images. We apply this elementary principle to the observation images and to the state image separately.

The orthogonal image has minimum separation

$$s_P = \frac{1}{2\sqrt{2}}.$$

If two observations have different first indices, their first two coordinates differ by at least $1/\sqrt{2}$, the chord length of one sixth-circle step after scaling. If their first indices agree, the distinct last coordinates are the four scaled values $1, 1/2, -1/2, -1$; their smallest spacing is $1/(2\sqrt{2})$. Thus an error strictly below $1/(4\sqrt{2})$ allows unique nearest-observation decoding on all 24 image points. The retained sheet then selects the admitted representative. This finite procedure includes the silhouette states, although continuous sheet lifting near the silhouette has a different conditioning problem.

For the paired phase image, an allowed nonzero index difference has coordinates of the same parity. If both differences are odd, their squared chord lengths sum to at least two. If an even difference is nonzero, that coordinate contributes at least three. The minimum separation is consequently

$$s_H = \sqrt{2},$$

attained by a simultaneous one-step change in both phases. Numerical error below $1/\sqrt{2}$ gives a unique nearest paired observation. The antipodal receipt remains necessary after this decoding: switching it changes the recovered geometric state by distance two while leaving the exact paired observation unchanged. No decrease of a continuous observation error can resolve an exact identification of this kind.

Finally the state image $E(D)$ has minimum chord separation $1/\sqrt{2}$. A change in one clock by one step attains this distance, and every nonzero coordinate change contributes at least $1/2$ to the squared distance of the scaled pair. Hence a reconstructed state with error below $1/(2\sqrt{2})$ has a unique nearest admitted state. This gives two related recovery routes: decode a finite observation and use its receipt, or perform a continuous lift within its valid domain and then decode the finite state. Their error constants arise from different maps and should be kept distinct.

6.5 A worked state and a deterministic error bound

Consider $(j, k) = (1, 2)$. Start from the odd-sector seed $(1, 0)$, apply the negative S_- step with parameter $-\theta$, and then apply S_+ with parameter θ. Their index increments are $(-1, +1)$ and $(1, +1)$, so the resulting change is $(0, 2)$. The exact observation and omitted coordinate are

$$x = \left(\frac{1}{2\sqrt{2}}, \frac{\sqrt{3}}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}}\right), \qquad w = \frac{\sqrt{3}}{2\sqrt{2}} > 0.$$

The orthogonal receipt is + and the integer code is 13. The paired phases are $(u, v) = (3, 5)$, for which (6.3) gives $j_0 = 1$, $k_0 = 2$ and receipt $b = 0$. The value thirteen follows from $1 + 6 \cdot 2$; no correspondence with a separately defined thirteen-count tape follows from this coincidence.

For a quantitative calculation, suppose each computed round operation maps S^3 to itself and differs from its exact action on the same input by chord error at most 10^{-4}. Suppose observation quantization adds Euclidean error at most 10^{-3}, and retain the correct sheet. These are synthetic deterministic error inputs. In particular, they are bounds rather than measured standard deviations or statements about the output of a physical apparatus.

The exact operations are isometries. Applying (5.2) to the two-step word gives final geometric error at most $2 \cdot 10^{-4}$. Orthogonal projection is nonexpansive, so the resulting observation $\widehat{x}$ satisfies

$$\|\widehat{x} - x\| \le \frac{1}{1000} + \frac{2}{10000} = \frac{3}{2500} = 0.0012. \tag{6.4}$$

Because $\|x\| = \sqrt{5/8}$ and $\sqrt{5/8} + 3/2500 < \sqrt{3}/2$, both observations lie in the closed ball of radius $\sqrt{3}/2$. The straight segment between them also lies there. Proposition 2.6 therefore applies with the conservative margin $a = 1/2$ and bounds the lifted-state error by $3/1250 = 0.0024$.

This last number is strictly below the state decoding radius $1/(2\sqrt{2})$. Nearest-state decoding recovers $(1, 2)$ exactly. We have therefore proved an exact conclusion about a finite state from inexact intermediate calculations. Its validity uses the operation bound, the observation bound, the sheet choice, and the admitted finite alphabet. An observation residual by itself would not supply all four hypotheses.

Corollary 6.1 (A neighborhood of exact finite recovery)**.** *For the state* $(1, 2)$*, retain the positive sheet and define*

$$\delta_* = \min\left\{\frac{\sqrt{3}}{2} - \sqrt{\frac{5}{8}}, \frac{1}{4\sqrt{2}}\right\} > 0.$$

Every observation y *with* $\|y - x\| < \delta_*$ *has a well-defined positive lift whose unique nearest admitted state is* $E(1, 2)$*.*

Proof. The first term in the minimum places y and x in the lift domain with margin $1/2$. Their lifted distance is therefore at most $2\|y - x\|$. The second term makes this distance strictly less than half the minimum state separation. Unique nearest-state decoding follows from the triangle inequality. □

This neighborhood expresses a local robustness property of the encoding and reader. Its radius is determined jointly by the distance to the chosen lift boundary and by the finite state spacing. The numerical inputs above produce one point inside this admissible neighborhood; the corollary describes all observations satisfying the same bound. A different sheet choice fails its hypothesis even if the projected numerical error is zero.

6.6 A calibrated period record

Attach to each canonical representative the cochain

$$h_{j,k} = (j, k, j + k)$$

on the torus complex (4.1). Its coordinates have units of dimensionless clock ticks, with one tick corresponding to angle θ. Since its third coordinate is the sum of the first two, it belongs to the harmonic space computed in Chapter 4. The period map $M(x, y, z) = (x, y)$ reads its two independent coordinates.

The use of representatives here is substantive. The map $D \to \mathbb{R}^3$ is a canonical phase record; crossing a clock wrap changes the printed representative. It is not a continuous real-valued angular coordinate on the whole torus. An accumulated winding record instead retains integer turns and depends on history. The present harmonic cochain records the admitted final state. Indeed its first two coordinates already determine that state, so this accompanying record constitutes additional retained information rather than an independent proof of an observation's injectivity.

For the worked state, introduce the raw record

$$\alpha = (1, 2, 3) + \frac{1}{1000}(1, 1, -1) = (1.001, 2.001, 2.999).$$

Writing $v = (1, 1, -1)$ gives $\rho = \langle \alpha, v \rangle = 3/1000$. The face residual is $(\rho, -\rho)$ and has norm $3\sqrt{2}/1000$. Orthogonal harmonic correction subtracts $\rho v/3$, returning $h = (1, 2, 3)$ exactly. The

removed component has norm $\sqrt{3}/1000$, equal to the spectral-gap bound of Proposition 5.1. Equality occurs because this complex has a single nonzero spectral direction, with eigenvalue six.

Read the two periods after this correction, and suppose their data are $m = (1, 2) + \xi$ with $\|\xi\| \le 10^{-4}$. Reconstruction takes the explicit form

$$\widehat{h} = (m_1, m_2, m_1 + m_2).$$

For $\xi = (\xi_1, \xi_2)$,

$$\left\|\widehat{h} - h\right\|^2 = \xi_1^2 + \xi_2^2 + (\xi_1 + \xi_2)^2 \le 3\,\|\xi\|^2\,,$$

so $\left\|\widehat{h} - h\right\| \le \sqrt{3}/10000$. The factor $\sqrt{3}$ agrees with the reciprocal harmonic response derived earlier. Reading the periods of the uncorrected α would retain the bias $(1/1000, 1/1000)$. Specifying the order of correction and measurement is therefore part of the reconstruction argument.

6.7 The combined reconstruction

Take the common observable to be

$$q(j, k) = (P(E(j, k)), h_{j,k}) \in \mathbb{R}^3 \oplus \mathbb{R}^3,$$

using the declared dimensionless coordinate norms. The product observation includes geometric data and the accompanying period record. The latter is injective on D, so the combined observable is injective as well. Agreement between its representations is a correspondence assertion for these specified maps; it is not evidence that the two records were acquired independently.

For the constructed example, (6.4) and the period estimate give, in the Euclidean product norm,

$$\left\|(\widehat{x}, \widehat{h}) - q(1, 2)\right\| \le \frac{3}{2500} + \frac{\sqrt{3}}{10000} < 0.001374.$$

The sum is a conservative upper bound on the product norm and requires no probabilistic independence assumption. Thus a common-observable allowance of 0.0014 and a lifted-state allowance of 0.0025 both contain the calculated errors. The same hypotheses also yield exact nearest-state decoding, as proved above.

The example completes a finite chain of correspondence. A pair of cyclic indices determines a polytope vertex and a round state; the selected generators act compatibly on both; each observation has an explicit fiber and a specified recovery datum; the harmonic record has a known local defect and a quantitatively stable two-period reconstruction. Every recovery claim can be traced to an inverse, a separation margin, or an operator bound. The continuous enclosure and its finite sector thereby support a single mathematical account without requiring their distinct observations to erase the same information.

Chapter 7

Enclosure and recoverability

7.1 The completed correspondence

The construction began with an enclosure and ends with a family of inverse problems. Its whole is explicit: $F = \partial(P_6 \times P_6)$, with a stated cell structure and a homeomorphism to S^3. Its round motions are explicit: left and right phase multiplication. Its observations are explicit: orthogonal projection, individual and paired Hopf maps, and period readers on a selected complex. Their fibers specify which information must accompany an observation if a representative is to be recovered.

The finite realization brings these statements into a single exact result.

Theorem 7.1 (Finite representation correspondence)**.** *Let $D = (\mathbb{Z}/6\mathbb{Z})^2$, with canonical representatives $0 \le j, k < 6$, and let E_F, E be the encodings of* (6.1)*. Each of the following is an injective description of D, with an explicit inverse on its image:*

1. *the integer code $j + 6k$;*
2. *the polytope vertex $E_F(j, k)$ and the round state $E(j, k)$;*
3. *the orthogonal observation with its prescribed sheet receipt;*
4. *the paired Hopf observation with its antipodal receipt;*
5. *the harmonic phase record $h_{j,k} = (j, k, j + k)$.*

The selected finite steps S_+, S_- intertwine with the round circle actions at parameter $\pi/3$. Every finite operation has an induced action on each of the five description images, obtained by encoding its action on D. These descriptions therefore carry the same admitted states and operations when their full retained data are used.

Proof. Division with remainder inverts the integer code. The two sixth roots of unity invert the geometric encoding, and Ψ^{-1} returns the polytope vertex. Orthogonal recovery uses the two-sheet formula, with the canonical singleton convention at the silhouette. Equation (6.3) and the retained bit invert the paired observation. Reading the first two coordinates inverts the harmonic record. All these formulas are established in Chapter 6. Multiplication by $e^{i\pi/3}$ in the specified complex factors proves the intertwining identity. If A is any one of these injective descriptions and $T : D \to D$ is a finite operation, its action on $A(D)$ is ATA^{-1}. This definition proves compatibility of the transported actions. □

The theorem concerns the augmented observations. Its third and fourth items deliberately retain different labels, because their unaugmented maps identify different states. It also concerns final configurations. The clock relations allow many operation words to reach the same state, so a history

reader needs further retained information. Finally, the harmonic record already contains the two phase indices. Its coexistence with the geometric observation provides a concrete comparison of descriptions, while statistical independence of records would be a separate acquisition premise.

7.2 Exact structure and quantitative response

There are two kinds of mathematical control in the result. The first is exact structure: the specified sphere homeomorphism, the group relations, the incidence identity $d_1 d_0 = 0$, and injectivity of the retained data. The second is quantitative response: the sheet margin, the finite separation distances, the spectral gap, and the smallest harmonic measurement response. Exact structure makes a recovery question well-defined. A positive response bound controls its behavior under perturbations.

These controls have complementary roles. A correct sheet with a small margin can amplify a small geometric discrepancy substantially. A wrong sheet can produce zero projected discrepancy and still recover a different state. Likewise, a cochain with zero local residual may have an arbitrary harmonic component until its periods are measured. The spectral gap controls the component orthogonal to that kernel; the harmonic response controls the remaining component. The reconstruction estimates of Chapter 5 make this division explicit.

For the worked state, the declared operation and observation errors produce an observation bound of 0.0012 and a same-sheet lift bound of 0.0024. The period perturbation contributes at most $\sqrt{3}/10000$. Their stated common-observable bound is less than 0.001374, and the lifted state lies strictly inside the unique-decoding radius. Thus continuous estimates lead to exact recovery of a finite state. The conclusion follows from the admitted alphabet and the margins as well as from the small residuals.

This also gives a precise meaning to consistency of descriptions in this model. Their exact diagrams commute, and their perturbed readings lie within bounds derived from the same specified maps. Such agreement is a result about a mathematical representation and its declared inputs. A physical application adds the task of establishing that those inputs and observation maps describe the apparatus at its calibrated resolution. The mathematical comparison can then be used as part of that application without changing the domain of its theorem.

7.3 Extensions of the construction

The product of two hexagons was chosen to make the twelve prism facets and the six-position clocks coincide exactly. Products of other polygons offer a natural extension. Their boundaries remain three-spheres after the corresponding factorwise disk construction. The arithmetic of the selected finite steps depends on the clock size: the same phase matrix is invertible modulo an odd size, whereas an even size retains an index-two obstruction. The twofold ambiguity of the continuous torus map still requires its own lift. Discrete sampling and continuous observation must therefore be compared for the chosen alphabet rather than inferred from one another.

Refining a phase alphabet illustrates why a limit requires its own response analysis. With N equally spaced positions in each factor and the same round normalization, the least state separation is

$$s_N = \sqrt{2}\sin(\pi/N).$$

A one-step change in either factor attains this distance, and every distinct pair has at least that separation. Unique nearest-state decoding is therefore ensured by a state error strictly below $\sin(\pi/N)/\sqrt{2}$. The radius tends to zero as the alphabet is refined. A continuous inverse estimate

can remain valid on a fixed sheet margin while the noise allowance for recovering an *exact discrete label* shrinks. The target of reconstruction must be specified when taking such a limit.

For nested admitted domains, extending the exact correspondence also requires compatible encodings and readers: encoding a state before inclusion must agree with including its encoding, and the observation diagrams must commute on the shared states. Uniform stability asks, in addition, which inverse constants remain bounded across the family. The selected six-position example makes these questions visible without assuming that a result for one finite domain has already supplied their limiting answer.

A second extension concerns richer measured complexes. The small torus calculation makes every incidence and response constant visible. Refinement introduces more cochain coordinates and requires compatible maps between complexes, controlled inner products, and estimates uniform in the chosen family. The transport argument shows what a pure change of coordinates preserves; a convergence argument for refinement would supply additional analytic control [1].

A physical realization would specify a measured quantity for each relevant coordinate, the apparatus that reports it, and the conversion between geometric or cochain discrepancy and measurement uncertainty [9, 8]. The enclosure would then participate in a measurement model with declared scales and a testable response. The present construction supplies the geometric and algebraic maps on which such a model can be built.

The central object throughout has been a whole accompanied by its recoverable observations. Enclosure fixes the state space; operations specify how its states change; observation fibers identify what is lost; retained lifting data make an inverse possible; and calibrated response bounds determine how reliably that inverse can be used. On the finite sector developed here, every passage is explicit.

Appendix A

Sources and the episode correspondence

A.1 Origin of the construction

The term *frinkahedron* and the questions about hidden enclosure, counting, and retained readings originate in the *Measurement* episode manuscripts. The root document `SPECIFICATION.md` sets out the geometric program. The earlier `instrument/` study develops the observation, transport, local correction, period, and information framework. The present monograph selects an explicit product polytope and finite state sector to make the geometric realization complete on a stated domain.

The construction contributes the vertices, prism facets, incidence relations, seam-preserving sphere map, circle actions, and thirty-six-state encoding used in the text. These are definitions followed by mathematical proofs. Their relation to the local source passages is recorded below so that a shared name does not substitute for a map between constructions.

A.2 Episode concordance

The sources were inspected on 21 October 2026. Paths in this section are relative to `device/Measurement/`. Line numbers locate the inspected working-tree version; the named declarations provide the more stable anchors.

A.2.1 The enclosing picture

`Episode03.lean`, lines 964–1005, discusses the hidden-hole picture and draws the frinkahedron. `Episode06.lean`, lines 37–50, lists twelve conceptual fenceposts. The twelve prism facets of Chapter 2 are the facets of the selected product polytope. The text proves their incidence structure directly. The episode passages motivate the enclosure question but do not themselves provide this inventory of polygon attachments.

A.2.2 Once and twice around

`Episode05.lean`, lines 67–68, declares the `once_around` and `twice_around` fields of type `SpaceTimePath`. In `Episode08.lean`, lines 983–1001, `once_around_the_integral`, `twice_around_the_integral`, and their named slack theorems concern integer counts and signs. The two circle actions in Chapter 3 are defined by their phase maps. An identification with a source path

datatype would require a state map and an intertwining proof for that datatype. The finite clock sector supplies such a proof for its own generators.

A.2.3 The retained tape reader

`Episode08.lean`, lines 1009–1025, defines `the_register` by a Horner fold and states the all-ones and monomial identities. This is an exact reader of a retained tape. The integer code $j + 6k$ of Chapter 6 is another exact reader with its own inverse. It reads two six-position indices. A comparison with the tape reader would specify a tape encoding of those indices and prove agreement after applying the selected readers.

A.2.4 A numerical receipt beside a quotient

`Episode05.lean`, lines 1137–1168, defines `TruthOrder` and `FACT.SAME`. `Episode08.lean`, lines 1148–1160, defines cross-multiplied `ReadingOrder`, `any_pair_is_its_inferred_integer`, and `theReadingCertified`. The last retains an integer together with a quotient equality. Proposition 4.2 explains the general mathematical distinction: a descended reader is constant on each quotient class, while an augmented output may retain a representative that the quotient alone has forgotten. The finite constant-quotient example exhibits this distinction explicitly.

A.3 The earlier observation framework

The `instrument/` sources are used through specific constructions. `AXIOMS.md` formulates exact observation equivalence and retained recovery. The distance chapter, `chapters/05.tex`, develops the scalar trace estimate and the two-projection commutator reproduced in Chapter 5. The transport chapter, `chapters/07.tex`, carries metrics and readers together. The measured loop framework in `FOURTH_ARGUMENT.md` and `chapters/08.tex` motivates harmonic recovery; the symmetry and entropy chapters develop compatible operations and the information lost by compression.

The present text provides its own proofs for the selected sphere, finite fiber counts, torus matrices, inverse estimates, and worked finite recovery. The scalar trace estimate is identified as an inherited result with its sampling and calibration assumptions retained. External references support the standard topology, Clifford, Hodge, approximation, and numerical-analysis frameworks. They do not replace the model-specific maps proved here.

A.4 Mathematical construction and source evidence

The exact established correspondence is among the finite clock states, their polytope and round encodings, their retained observations, and their period records. Theorem 7.1 states that result. The entire episode datatype and a physical spacetime interpretation have larger domains than this finite construction. Extending the correspondence to either would require the relevant encoding, operations, observation maps, and calibration.

The repository records inspected source identities in `pdfs/SOURCE_SNAPSHOT.md`. Its hashes identify working-tree bytes; they are not a report of a Lean build. The independent finite calculation in `pdfs/specification/scripts/check_model.py` reproduces the stated finite observations and matrix identities. The worked error allowances in Chapter 6 are synthetic deterministic inputs, clearly separated from an apparatus measurement. This provenance keeps the proof of the selected model and the evidence for any later application available for independent examination.

Bibliography

[1] Douglas N. Arnold, Richard S. Falk, and Ragnar Winther. Finite element exterior calculus: from Hodge theory to numerical stability. *Bulletin of the American Mathematical Society*, 47(2):281–354, 2010. Author version: arXiv:0906.4325.

[2] Carl de Boor. *A Practical Guide to Splines*. Springer, New York, 1978.

[3] Carl de Boor and Robert E. Lynch. On splines and their minimum properties. *Journal of Mathematics and Mechanics*, 15(6):953–969, 1966.

[4] Jean Gallier. Clifford algebras, clifford groups, and a generalization of the quaternions, 2008. arXiv:0805.0311; revised 2014.

[5] Gene H. Golub and Charles F. Van Loan. *Matrix Computations*. Johns Hopkins University Press, 4 edition, 2013.

[6] Allen Hatcher. *Algebraic Topology*. Cambridge University Press, 2002.

[7] Xiaoye Jiang, Lek-Heng Lim, Yuan Yao, and Yinyu Ye. Statistical ranking and combinatorial Hodge theory, 2008. arXiv:0811.1067; revised 2009.

[8] Joint Committee for Guides in Metrology. Evaluation of measurement data: Guide to the expression of uncertainty in measurement. Technical Report JCGM 100:2008, Bureau International des Poids et Mesures, 2008.

[9] Joint Committee for Guides in Metrology. Guide to the expression of uncertainty in measurement: Part 6: Developing and using measurement models. Technical Report JCGM GUM-6:2020, Bureau International des Poids et Mesures, 2020.

[10] David W. Lyons. An elementary introduction to the Hopf fibration. *Mathematics Magazine*, 76(2):87–98, 2003.

[11] Saul Schleimer and Henry Segerman. Sculptures in S^3. In *Proceedings of Bridges 2012: Mathematics, Music, Art, Architecture, Culture*, pages 103–110, 2012.

[12] Peter Woit. Clifford algebras and spin groups. Columbia University, Math G4344 lecture notes, 2012. Spring 2012.